A Project Report On

Power Grid Failure Detection Based on Sensing Frequency and Voltage Variation using IoT

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CERTIFICATE

This is to certify that the project report entitled **Power Grid Failure Detection Based on Sensing Frequency and Voltage Variation using IoT** submitted by **Thodupunuri Saneeth (2451-20-734-006), Vankdoth Sai Tarun (2451-20-734-019), Mohammed Abdul Raheem Saad (2451-20-734-034)** to the department of EEE, Maturi Venkata Subba Rao (MVSR) Engineering College, in the partial fulfillment of the requirement for the degree of Bachelor of Engineering in Electrical and Electronics Engineering is a bona fide record of project work carried out under my supervision. The contents of this report, in full or in parts, have not been submitted to any other Institution or University for the award of any degree or diploma.

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Signature of External Examiner

DECLARATION

We declare that this project report titled **Power Grid Failure Detection Based on Sensing Frequency and Voltage Variation using IoT** submitted in partial fulfillment of the degree of Bachelor of Engineering in Electrical and Electronics Engineering is a record of original work carried out by us under the supervision of **S.Shyam Mohan**, and has not formed the basis for the award of any other degree or diploma, in this or any other Institution or University. In keeping with the ethical practice in reporting scientific information, due acknowledgements have been made wherever the findings of others have been cited.

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ABSTRACT

Grid synchronization failure can cause complete black out. So, there is always a need for a system that can sense any abnormalities and take actions accordingly to prevent black outs. A grid is connected with several power generating units like thermal, nuclear, wind etc. to deliver power to the load. In India, the generating units have to deliver power according to the Indian Electricity Grid Code, 2010 which states that the variation in voltage should be within the limit of ±10% and that for the frequency should be within ±3%. If any value higher/lower is detected, then that particular feeder should be disconnected from the grid temporarily in order to protect the grid. In this paper, we will be discussing about a system based on the microcontroller, that will be used to monitor the variation in voltage and frequency of the any external supply source, and automatically disconnects the supply source from the load.

This system is based on a microcontroller of PIC16F76/73. The microcontroller monitors the under/over voltage being derived from a set of comparators. As the frequency of the mains supply cannot be changed, the project uses comparator for changing the frequency, while a rectifier is used to vary the input voltage to test the functioning of the project. A lamp load (indicating a predictable blackout, brownout) being driven from the microcontroller in case of voltage/frequency going out of acceptable range.

The design of a system to detect the synchronization failure of any external supply source to the power grid on sensing the abnormalities in frequency and voltage and thereby protecting the load. There are several power generation units connected to the grid such as hydra, thermal, solar etc., to supply power to the load. These generating units need to supply power according to the rules of the grid. These rules involve maintaining a voltage variation within limits and also the frequency. If any deviation from the acceptable limit of the grid, it is mandatory that the same feeder should automatically get disconnected from the grid which by effect is termed as islanding. This prevents in large scale brown out or black out of the grid power. So, it is preferable to have a system which can warn the grid in advance so that alternate arrangements are kept on standby to avoid complete grid failure. This system is based on Arduino Uno microcontroller. The microcontroller monitors the under/over voltage being derived from a set of comparators and a standard Arduino is used to vary the input voltage to test the functioning of the project. A lamp load (indicating a predictable blackout, brownout) being driven from the microcontroller in case of voltage/frequency going out of acceptable range.

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ABBREVIATIONS

PIC	PERIPHERAL INTERFACE CONTROLLER
LCD	LIQUID CRYSTAL DISPLAY
PC	PERSONAL COMPUTER
DSP	DIGITAL SIGNAL PROCESSORS
RAM	RANDOM ACCESS MEMORY
ROM	READ ONLY MEMORY
CPU	CENTRAL PROCESSING UNIT
FPU	FLOATING-POINT PROCESSING UNIT
UI	USER INTERFACE
FPGA	FIELD PROGRAMMABLE GATE ARRAY
ADC	ANALOG TO DIGITAL CONVERTER
POT	POTENTIOMETER
LED	LIGHT EMITTING DIODE
PWRT	POWER –UP TIMER
OST	OSCILLATOR START-UP TIMER
WDT	WATCHDOG TIMER
ICSP	IN-CIRCUIT SERIAL PROGRAMMING
RISC	REDUCED INSTRUCTION SET COMPUTER
CISC	COMPLEX INSTRUCTION SET COMPUTER
DIP	DUAL IN PACKAGE
PCB	PRINTED CIRCUIT BOARD
SSR	SOLID STATE RELAY
IoT	INTERNET OF THINGS
RPS	REGULATED POWER SUPPLY

CHAPTER 1: INTRODUCTION

1.1 Introduction:

The purpose of this project is to monitor and control various electrical devices within the substation. The parameters that will be monitored include the voltage, frequency and circuit breaker status. It also trips the circuit breaker whenever necessary. This system provides both audible and visual alerts using LCD.

This project consists of a Microcontroller based control system, sensors for various electrical parameters and sensor for voltage and frequency measurement. The analog and digital inputs from respective sensors are sent to the microcontroller. Microcontroller process this raw data and converts into human readable format and displays on LCD display. Microcontroller continuously checks the data being read and takes necessary action if the read values exceed the preset limits. The action could be opening of a circuit breaker.

1.2 Project Overview:

An embedded system is a combination of software and hardware to perform a dedicated task. Some of the main devices used in embedded products are Microprocessors and Microcontrollers.

Microprocessors are commonly referred to as general purpose processors as they simply accept the inputs, process it and give the output. In contrast, a microcontroller not only accepts the data as inputs but also manipulates it, interfaces the data with various devices, controls the data and thus finally gives the result.

The project **Power Grid Failure detection based on sensing frequency and voltage variation using IoT** using PIC16F73 Microcontroller is an exclusive project that can control and monitor the devices according to the instructions given by the above said microcontroller.

1.3 Thesis Overview:

The thesis explains the implementation of **Power Grid Failure detection based on sensing frequency and voltage variation using IoT** using 16F73 microcontroller. The organization of the thesis is explained here with:

Chapter 1 Presents introduction to the overall thesis and the overview of the project. In the project overview a brief introduction of **Power Grid Failure detection based on sensing frequency and voltage variation using IoT** and its applications are discussed.

Chapter 2 Presents the topic embedded systems. It explains the about what is embedded systems, need for embedded systems, explanation of it along with its applications.

Chapter 3 Presents the hardware description. It deals with the block diagram of the project and explains the purpose of each block. In the same chapter the explanation of microcontrollers, power supplies, voltage sensor, potentiometer, frequency measuring unit, relay, LCD, are considered.

Chapter 4 Presents the software description. It explains the implementation of the project using PIC C Compiler software.

Chapter 5 Presents the project description along with LCD, buzzer, relay, potentiometer module interfacing to microcontroller.

Chapter 6 Presents the advantages, disadvantages and applications of the project.

Chapter 7 Presents the results, conclusion and future scope of the project.

CHAPTER 2: EMBEDDED SYSTEMS

2.1 Embedded Systems:

An embedded system is a computer system designed to perform one or a few dedicated functions often with real-time computing constraints. It is embedded as part of a complete device often including hardware and mechanical parts. By contrast, a general-purpose computer, such as a personal computer (PC), is designed to be flexible and to meet a wide range of end-user needs. Embedded systems control many devices in common use today.

Embedded systems are controlled by one or more main processing cores that are typically either microcontrollers or digital signal processors (DSP). The key characteristic, however, is being dedicated to handle a particular task, which may require very powerful processors. For example, air traffic control systems may usefully be viewed as embedded, even though they involve mainframe computers and dedicated regional and national networks between airports and radar sites. (Each radar probably includes one or more embedded systems of its own.)

Since the embedded system is dedicated to specific tasks, design engineers can optimize it to reduce the size and cost of the product and increase the reliability and performance. Some embedded systems are mass-produced, benefiting from economies of scale.

Physically embedded systems range from portable devices such as digital watches and MP3 players, to large stationary installations like traffic lights, factory controllers, or the systems controlling nuclear power plants. Complexity varies from low, with a single microcontroller chip, to very high with multiple units, peripherals and networks mounted inside a large chassis or enclosure.

In general, "embedded system" is not a strictly definable term, as most systems have some element of extensibility or programmability. For example, handheld computers share some elements with embedded systems such as the operating systems and microprocessors which power them, but they allow different applications to be loaded and peripherals to be connected. Moreover, even systems which don't expose programmability as a primary feature generally need to support software updates. On a continuum from "general purpose" to "embedded", large application systems will have subcomponents at most points even if the system as a whole is

"designed to perform one or a few dedicated functions", and is thus appropriate to call "embedded". A modern example of embedded system is shown in fig: 2.1.

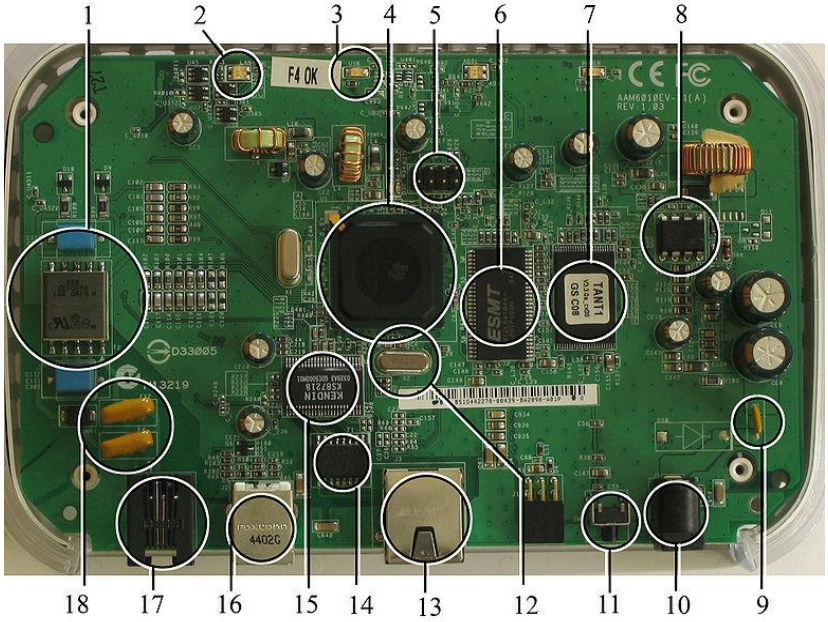


Fig 2. 1: A modern example of embedded system

Labeled parts include microprocessor (4), RAM (6), flash memory (7).Embedded systems programming is not like normal PC programming. In many ways, programming for an embedded system is like programming PC 15 years ago. The hardware for the system is usually chosen to make the device as cheap as possible. Spending an extra dollar a unit in order to make things easier to program can cost millions. Hiring a programmer for an extra month is cheap in comparison. This means the programmer must make do with slow processors and low memory, while at the same time battling a need for efficiency not seen in most PC applications. Below is a list of issues specific to the embedded field.

2.1.1 Resources:

To save costs, embedded systems frequently have the cheapest processors that can do the job. This means your programs need to be written as efficiently as possible. When dealing with large data sets, issues like memory cache misses that never matter in PC programming can hurt you. Luckily, this won't happen too often- use reasonably efficient algorithms to start, and optimize only when necessary. Of course, normal profilers won't work well, due to the same reason debuggers don't work well.

Memory is also an issue. For the same cost savings reasons, embedded systems usually have the least memory they can get away with. That means their algorithms must be

memory efficient (unlike in PC programs, you will frequently sacrifice processor time for memory, rather than the reverse). It also means you can't afford to leak memory. Embedded applications generally use deterministic memory techniques and avoid the default "new" and "malloc" functions, so that leaks can be found and eliminated more easily. Other resources programmers expect may not even exist. For example, most embedded processors do not have hardware FPUs (Floating-Point Processing Unit). These resources either need to be emulated in software, or avoided altogether.

2.1.2 Real Time Issues:

Embedded systems frequently control hardware, and must be able to respond to them in real time. Failure to do so could cause inaccuracy in measurements, or even damage hardware such as motors. This is made even more difficult by the lack of resources available. Almost all embedded systems need to be able to prioritize some tasks over others, and to be able to put off/skip low priority tasks such as UI in favor of high priority tasks like hardware control.

2.2 Need For Embedded Systems:

The uses of embedded systems are virtually limitless, because every day new products are introduced to the market that utilizes embedded computers in novel ways. In recent years, hardware such as microprocessors, microcontrollers, and FPGA chips have become much cheaper. So when implementing a new form of control, it's wiser to just buy the generic chip and write your own custom software for it. Producing a custom-made chip to handle a particular task or set of tasks costs far more time and money. Many embedded computers even come with extensive libraries, so that "writing your own software" becomes a very trivial task indeed. From an implementation viewpoint, there is a major difference between a computer and an embedded system. Embedded systems are often required to provide Real-Time response. The main elements that make embedded systems unique are its reliability and ease in debugging.

2.3 APPLICATIONS OF EMBEDDED SYSTEMS:

2.3.1 Consumer applications:

At home we use a number of embedded systems which include microwave oven, remote control, vcd players, dvd players, camera etc….

Fig 2. 2: Automatic coffee makes equipment

2.3.2 Office automation:

We use systems like fax machine, modem, printer etc…

Fig 2. 3: Fax machine

Fig 2.4: Printing machine

2.3.3. Industrial automation:

Today a lot of industries are using embedded systems for process control. In industries we design the embedded systems to perform a specific operation like monitoring temperature, pressure, humidity ,voltage, current etc.., and basing on these monitored levels we do control other devices, we can send information to a centralized monitoring station.

Fig 2. 5: Robot

In critical industries where human presence is avoided there we can use robots which are programmed to do a specific operation.

2.3.4 Computer networking:

Embedded systems are used as bridges routers etc..

Fig 2. 6:Computer networking

CHAPTER 3: HARDWARE DESCRIPTION

3.1 Introduction:

In this chapter the block diagram of the project and design aspect of independent modules are considered. Block diagram is shown in fig: 3.1:

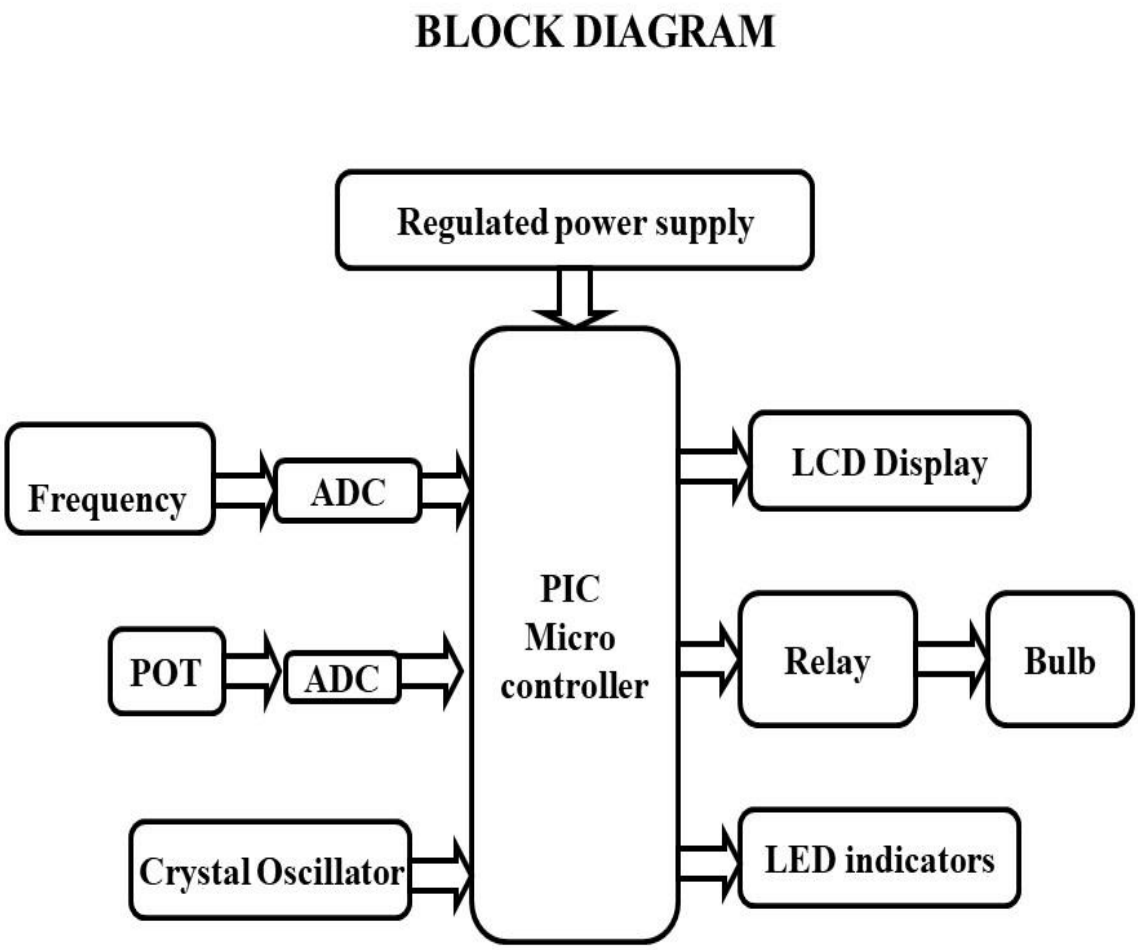


Fig 3.1 Block Diagram

The main blocks of this project are:

- ✓ Regulated power supply.
- ✓ PIC Microcontroller.
- ✓ Relay.
- ✓ LCD
- ✓ Opto coupler(frequency)
- ✓ Pot
- ✓ Crystal oscillator
- ✓ Reset.
- ✓ LED indicators

3.2 Micro controller:

3.2.1 Introduction to Microcontrollers:

Circumstances that we find ourselves in today in the field of microcontrollers had their beginnings in the development of technology of integrated circuits. This development has made it possible to store hundreds of thousands of transistors into one chip. That was a prerequisite for production of microprocessors, and the first computers were made by adding external peripherals such as memory, input-output lines, timers and other. Further increasing of the volume of the package resulted in creation of integrated circuits. These integrated circuits contained both processor and peripherals. That is how the first chip containing a microcomputer, or what would later be known as a microcontroller came about.

Microprocessors and microcontrollers are widely used in embedded systems products. Microcontroller is a programmable device. A microcontroller has a CPU in addition to a fixed amount of RAM, ROM, I/O ports and a timer embedded all on a single chip. The fixed amount of on-chip ROM, RAM and number of I/O ports in microcontrollers makes them ideal for many applications in which cost and space are critical.

The microcontroller used in this project is PIC16F73. The PIC families of microcontrollers are developed by Microchip Technology Inc. Currently they are some of the most

popular microcontrollers, selling over 120 million devices each year. There are basically four families of PIC microcontrollers:

PIC12CXXX 12/14-bit program word

PIC 16C5X 12-bit program word

PIC16CXXX and PIC16FXXX 14-bit program word

PIC17CXXX and PIC18CXXX 16-bit program word

The features, pin description of the microcontroller used are discussed in the following sections.

3.2.2 Description:

The PIC16F73 CMOS FLASH-based 8-bit microcontroller is upward compatible with the PIC16C73B/74B/76/77, PIC16F873/874/876/877devices. It features 200 ns instruction execution, self programming, an ICD, 2 Comparators, 8 channels of 8-bit Analog-to-Digital (A/D) converter, 2 capture/compare/PWM functions, a synchronous serial port that can be configured as either 3-wire SPI or 2-wire I2C bus, a USART, and a Parallel Slave Port.

Special Microcontroller Features

• Power-up Timer (PWRT) and oscillator Start-up Timer (OST)

• Watchdog Timer (WDT) with its own on-chip RC oscillator for reliable operation

• Programmable code protection

• Power saving SLEEP mode

• Selectable oscillator options

• In-Circuit Serial Programming (ICSP) via two Pins

Analog Comparator module

- 2 analog comparators
- Programmable on-chip voltage reference module
- Programmable input multiplexing from device inputs and internal VREF
- Comparator outputs are externally accessible

CMOS Technology:

• Low power, high speed CMOS FLASH technology

• Fully static design

• Wide operating voltage range: 2.0V to 5.5V

• High Sink/Source Current: 25 mA

• Industrial temperature range

• Low power consumption:

- < 2 mA typical @ 5V, 4 MHz

- 20 μA typical @ 3V, 32 kHz

- < 1 μA typical standby current

Following are the major blocks of PIC Microcontroller.

Program memory (FLASH) is used for storing a written program. Since memory made in FLASH technology can be programmed and cleared more than once, it makes this microcontroller suitable for device development.

EEPROM - data memory that needs to be saved when there is no supply. It is usually used for storing important data that must not be lost if power supply suddenly stops. For instance, one such data is an assigned temperature in temperature regulators. If during a loss of power supply this data was lost, we would have to make the adjustment once again upon return of supply. Thus our device losses on self-reliance.

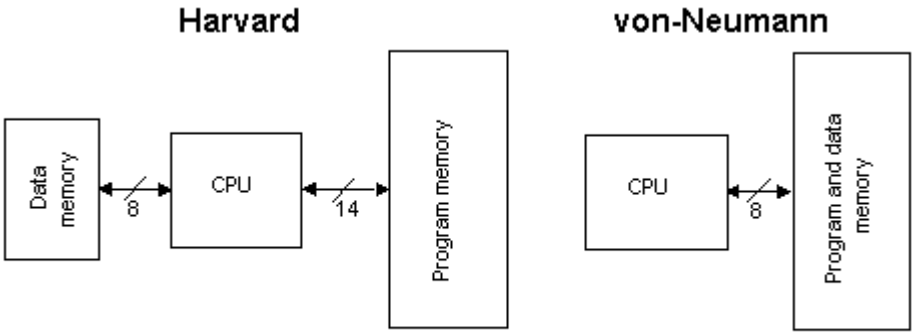
RAM - Data memory used by a program during its execution.

In RAM are stored all inter-results or temporary data during run-time.

PORTS are physical connections between the microcontroller and the outside world. PIC16F73 has 22 I/O.

FREE-RUN TIMER is an 8-bit register inside a microcontroller that works independently of the program. On every fourth clock of the oscillator it increments its value until it reaches the maximum (255), and then it starts counting over again from zero. As we know the exact timing between each two increments of the timer contents, timer can be used for measuring time which is very useful with some devices.

CENTRAL PROCESSING UNIT has a role of connective element between other blocks in the microcontroller. It coordinates the work of other blocks and executes the user program.



Harvard vs. von Neuman Block Architectures

CISC, RISC

It has already been said that PIC16F73 has RISC architecture. This term is often found in computer literature, and it needs to be explained here in more detail. Harvard architecture is a newer concept than von-Neumann. It rose out of the need to speed up the work of a microcontroller. In Harvard architecture, data bus and address bus are separate. Thus a greater flow of data is possible through the central processing unit, and of course, a greater speed of work. Separating a program from data memory makes it further possible for instructions not to have to be 8-bit words. PIC16F73 uses 14 bits for instructions which allows for all instructions to be one word instructions. It is also typical for Harvard architecture to have fewer instructions than von-Neumann's, and to have instructions usually executed in one cycle.

Microcontrollers with Harvard architecture are also called "RISC microcontrollers". RISC stands for Reduced Instruction Set Computer. Microcontrollers with von-Neumann's architecture are called 'CISC microcontrollers'. Title CISC stands for Complex Instruction Set Computer

Crystal oscillator:

The crystal oscillator speed that can be connected to the PIC microcontroller range from DC to 20Mhz. Using the CCS C compiler normally 20Mhz oscillator will be used and the price is very cheap. The 20 MHz crystal oscillator should be connected with about 22pF capacitor. Please refer to my circuit schematic.

There are 5 input/output ports on PIC microcontroller namely port A, port B, port C, port D and port E. Each port has different function. Most of them can be used as I/O port.

Applications

PIC16F73 perfectly fits many uses, from automotive industries and controlling home appliances to industrial instruments, remote sensors, electrical door locks and safety devices. It is also ideal for smart cards as well as for battery supplied devices because of its low consumption.

EEPROM memory makes it easier to apply microcontrollers to devices where permanent storage of various parameters is needed (codes for transmitters, motor speed, receiver frequencies, etc.). Low cost, low consumption, easy handling and flexibility make PIC16F73 applicable even in areas where microcontrollers had not previously been considered (example: timer functions, interface replacement in larger systems, coprocessor applications, etc.).

In System Programmability of this chip (along with using only two pins in data transfer) makes possible the flexibility of a product, after assembling and testing have been completed. This capability can be used to create assembly-line production, to store calibration data available only after final testing, or it can be used to improve programs on finished products.

Pin description

PIC16F73 has a total of 28 pins. It is most frequently found in a DIP28 type of case but can also be found in SMD case which is smaller from a DIP. DIP is an abbreviation for Dual In Package. SMD is an abbreviation for Surface Mount Devices suggesting that holes for pins to go through when mounting aren't necessary in soldering this type of a component.

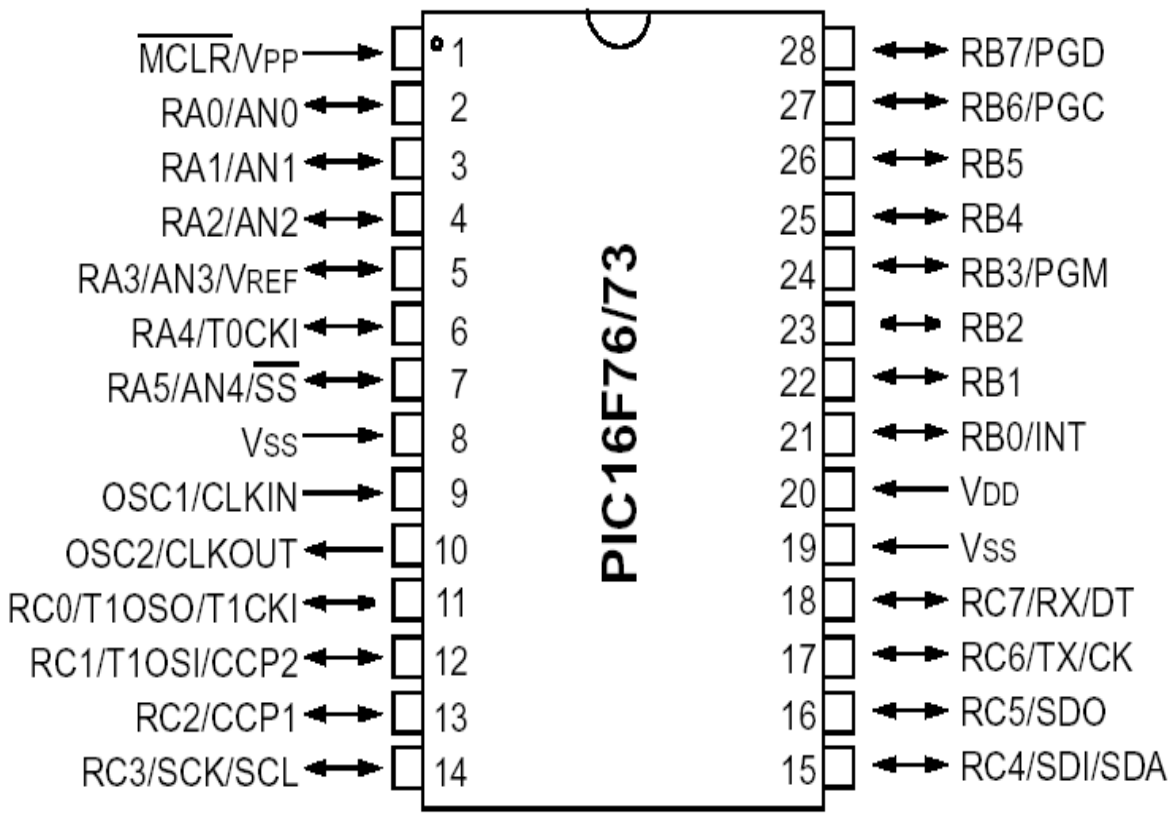


Fig 3. 2: Pin Description of Microcontroller

Pins on PIC16F73 microcontroller have the following meaning:

There are 28 pins on PIC16F73. Most of them can be used as an IO pin. Others are already for specific functions. These are the pin functions.

1. MCLR – to reset the PIC

2. RA0 – port A pin 0

3. RA1 – port A pin 1

4. RA2 – port A pin 2

5. RA3 – port A pin 3

6. RA4 – port A pin 4

7. RA5 – port A pin 5

8. VSS – ground

9. OSC1 – connect to oscillator

10. OSC2 – connect to oscillator

11. RC0 – port C pin 0 VDD – power supply

12. RC1 – port C pin 1

13. RC2 – port C pin 2

14. RC3 – port C pin 3

15. RC4 - port C pin 4

16. RC5 - port C pin 5

17. RC6 - port C pin 6

18. RC7 - port C pin 7

19. VSS – ground

20. VDD – power supply

21. RB0 - port B pin 0

22. RB1 - port B pin 1

23. RB2 - port B pin 2

24. RB3 - port B pin 3

25. RB4 - port B pin 4

26. RB5 - port B pin 5

27. RB6 - port B pin 6

28. RB7 - port B pin 7

By utilizing all of this pin so many application can be done such as:

1. LCD – connect to Port B pin.

2. LED – connect to any pin declared as output.

3. Relay and Motor - connect to any pin declared as output.

4. External EEPROM – connect to I2C interface pin – RC3 and RC4 (SCL and SDA)

5. LDR, Potentiometer and sensor – connect to analogue input pin such as RA0.

6. GSM modem dial up modem – connect to RC6 and RC7 – the serial communication interface using RS232 protocol.

For more detail function for each specific pin please refer to the device datasheet from Microchip.

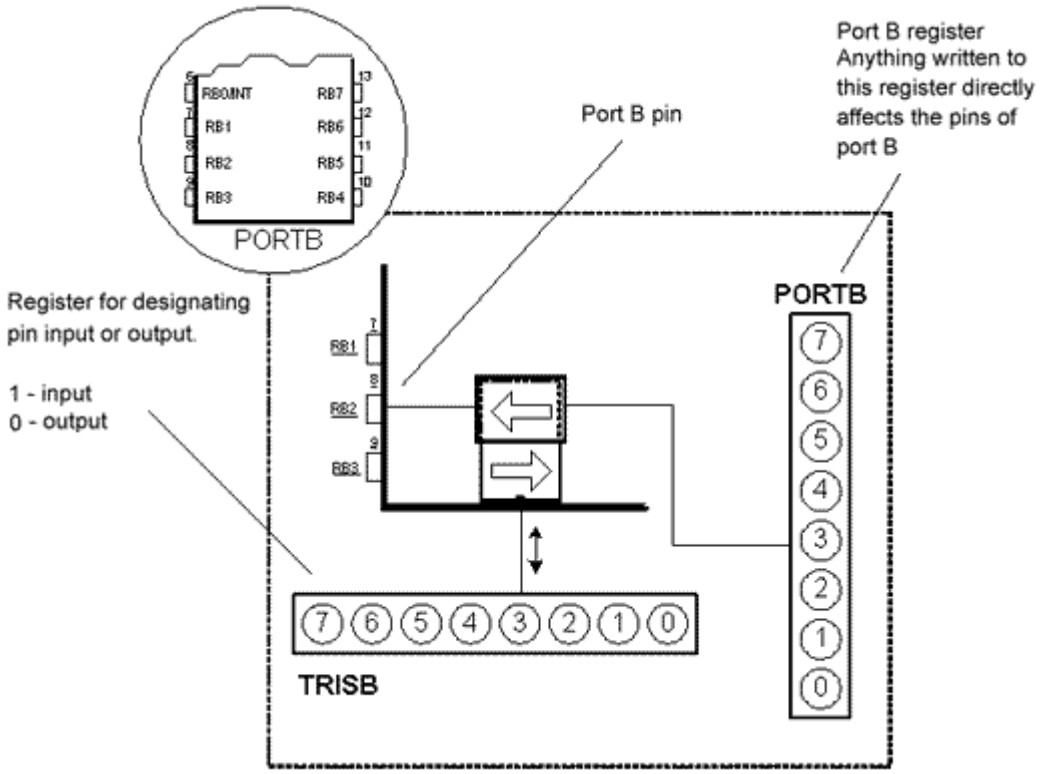
Ports

Term "port" refers to a group of pins on a microcontroller which can be accessed simultaneously, or on which we can set the desired combination of zeros and ones, or read from them an existing status. Physically, port is a register inside a microcontroller which is connected by wires to the pins of a microcontroller. Ports represent physical connection of Central Processing Unit with an outside world. Microcontroller uses them in order to monitor or control other components or devices. Due to functionality, some pins have twofold roles like PA4/TOCKI for instance, which is in the same time the fourth bit of port A and an external input for free-run counter. Selection of one of these two pin functions is done in one of the configuration registers. An illustration of this is the fifth bit T0CS in OPTION register. By selecting one of the functions the other one is disabled.

All port pins can be designated as input or output, according to the needs of a device that's being developed. In order to define a pin as input or output pin, the right combination of zeros and ones must be written in TRIS register. If the appropriate bit of TRIS register contains logical "1", then that pin is an input pin, and if the opposite is true, it's an output pin. Every port has its proper TRIS register. Thus, port A has TRISA, and port B has TRISB. Pin direction can be changed during the course of work which is particularly fitting for one-line communication where data flow constantly changes direction. PORTA and PORTB state registers are located in bank 0, while TRISA and TRISB pin direction registers are located in bank 1.

PORTB and TRISB

PORTB has adjoined 8 pins. The appropriate register for data direction is TRISB. Setting a bit in TRISB register defines the corresponding port pin as input, and resetting a bit in TRISB register defines the corresponding port pin as output.



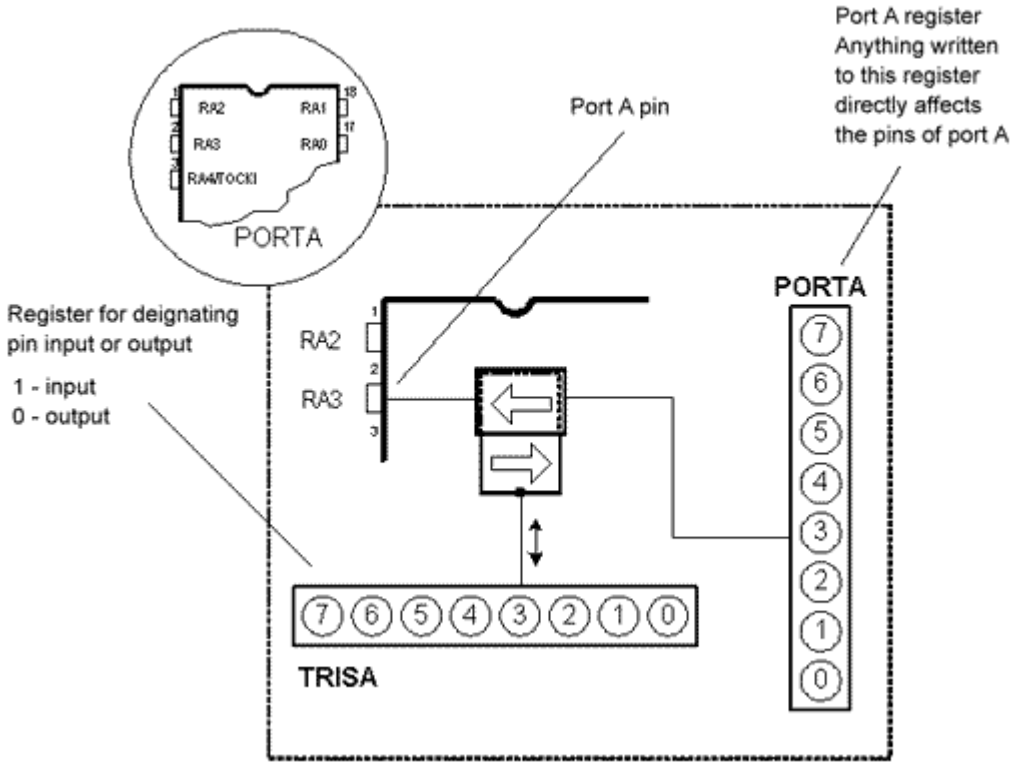
Each PORTB pin has a weak internal pull-up resistor (resistor which defines a line to logic one) which can be activated by resetting the seventh bit RBPU in OPTION register. These 'pull-up' resistors are automatically being turned off when port pin is configured as an output. When a microcontroller is started, pull-ups are disabled.

Four pins PORTB, RB7:RB4 can cause an interrupt which occurs when their status changes from logical one into logical zero and opposite. Only pins configured as input can cause this interrupt to occur (if any RB7:RB4 pin is configured as an output, an interrupt won't be generated at the change of status.) This interrupt option along with internal pull-up resistors makes it easier to solve common problems we find in practice like for instance that of matrix keyboard. If rows on the keyboard are connected to these pins, each push on a key will then cause an interrupt. A microcontroller will determine which key is at hand while processing an interrupt It is not recommended to refer to port B at the same time that interrupt is being processed

PORTA and TRISA

PORTA has 5 adjoining pins. The corresponding register for data direction is TRISA at address 85h. Like with port B, setting a bit in TRISA register defines also the corresponding port pin as input, and clearing a bit in TRISA register defines the corresponding port pin as output.

It is important to note that PORTA pin RA4 can be input only. On that pin is also situated an external input for timer TMR0. Whether RA4 will be a standard input or an input for a counter depends on T0CS bit (*TMR0 Clock Source Select bit*). This pin enables the timer TMR0 to increment either from internal oscillator or via external impulses on RA4/T0CKI pin.



Memory organization

PIC16F73 has two separate memory blocks, one for data and the other for program. EEPROM memory with GPR and SFR registers in RAM memory make up the data block, while FLASH memory makes up the program block.

Program memory

Program memory has been carried out in FLASH technology which makes it possible to program a microcontroller many times before it's installed into a device, and even after its

installment if eventual changes in program or process parameters should occur. The size of program memory is 1024 locations with 14 bits width where locations zero and four are reserved for reset and interrupt vector.

Data memory

Data memory consists of EEPROM and RAM memories. EEPROM memory consists of 256 eight bit locations whose contents are not lost during loosing of power supply. EEPROM is not directly addressable, but is accessed indirectly through EEADR and EEDATA registers. As EEPROM memory usually serves for storing important parameters (for example, of a given temperature in temperature regulators) , there is a strict procedure for writing in EEPROM which must be followed in order to avoid accidental writing. RAM memory for data occupies space on a memory map from location 0x0C to 0x4F which comes to 68 locations. Locations of RAM memory are also called GPR registers which is an abbreviation for *General Purpose Registers*. GPR registers can be accessed regardless of which bank is selected at the moment.

3.3 REGULATED POWER SUPPLY:

3.3.1 Introduction:

Power supply is a supply of electrical power. A device or system that supplies electrical or other types of energy to an output load or group of loads is called a power supply unit or PSU. The term is most commonly applied to electrical energy supplies, less often to mechanical ones, and rarely to others.

A power supply may include a power distribution system as well as primary or secondary sources of energy such as

- Conversion of one form of electrical power to another desired form and voltage, typically involving converting AC line voltage to a well-regulated lower-voltage DC for electronic devices. Low voltage, low power DC power supply units are commonly integrated with the devices they supply, such as computers and household electronics.
- Batteries.
- Chemical fuel cells and other forms of energy storage systems.
- Solar power.
- Generators or alternators.

3.3 Block Diagram:

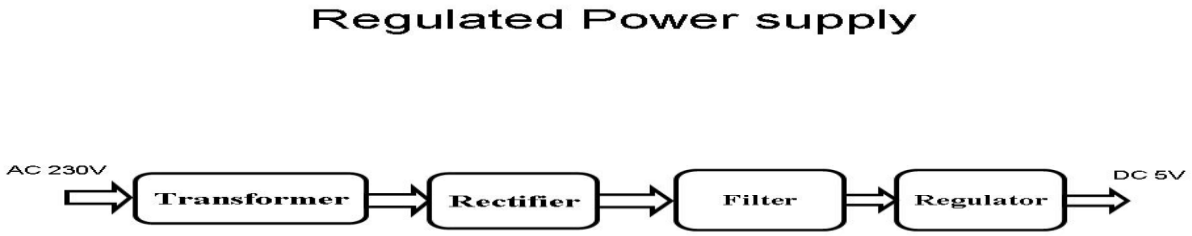


Fig 3.3: Regulated Power Supply

The basic circuit diagram of a regulated power supply (DC O/P) with LED connected as load is shown in fig: 3.4

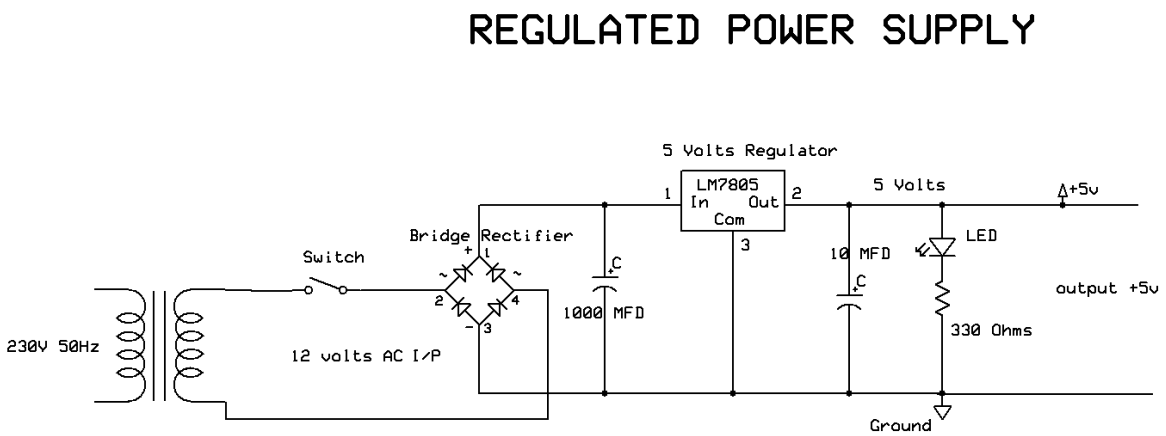


Fig 3.4: Circuit diagram of Regulated Power Supply with LED connection

The components mainly used in above figure are

- 230V AC MAINS
- TRANSFORMER
- BRIDGE RECTIFIER(DIODES)
- CAPACITOR

- VOLTAGE REGULATOR(IC 7805)
- RESISTOR
- LED(LIGHT EMITTING DIODE)

  The detailed explanation of each and every component mentioned above is as follows:

Step 1: Transformation: The process of transforming energy from one device to another is called transformation. For transforming energy we use transformers.

Transformers:

A transformer is a device that transfers electrical energy from one circuit to another through inductively coupled conductors without changing its frequency. A varying current in the first or primary winding creates a varying magnetic flux in the transformer's core, and thus a varying magnetic field through the secondary winding. This varying magnetic field induces a varying electromotive force (EMF) or "voltage" in the secondary winding. This effect is called mutual induction.

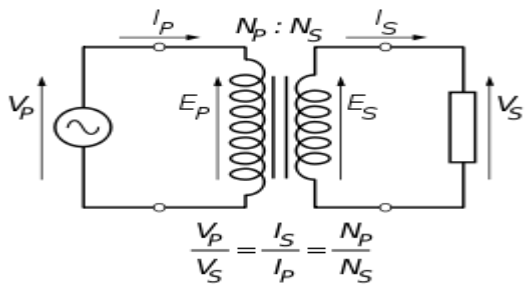


Fig 3.5: Step-Down Transformer

The voltage induced in the secondary is determined by the TURNS RATIO.

$$\frac{\text{primary voltage}}{\text{secondary voltage}} = \frac{\text{number of primary turns}}{\text{number of secondary turns}}$$

We can use the transformers as step up or step down.

Step Up transformer:

In case of step up transformer, primary windings are every less compared to secondary winding. Because of having more turns secondary winding accepts more energy, and it releases more voltage at the output side.

Step down transformer:

In case of step down transformer, Primary winding induces more flux than the secondary winding, and secondary winding is having less number of turns because of that it accepts less number of flux, and releases less amount of voltage.

Battery power supply:

A battery is a type of linear power supply that offers benefits that traditional line-operated power supplies lack: mobility, portability and reliability. A battery consists of multiple electrochemical cells connected to provide the voltage desired. Fig: 3.6 shows Hi-Watt 9V battery

Fig 3.6: Hi-Watt 9V Battery

Step 2: Rectification

The process of converting an alternating current to a pulsating direct current is called as rectification. For rectification purpose we use rectifiers.

Rectifiers:

A rectifier is an electrical device that converts alternating current (AC) to direct current (DC), a process known as rectification. Rectifiers have many uses including as components

of power supplies and as detectors of radio signals. Rectifiers may be made of solid-state diodes, vacuum tube diodes, mercury arc valves, and other components.

A device that it can perform the opposite function (converting DC to AC) is known as an inverter.

DB107:

Now -a -days Bridge rectifier is available in IC with a number of DB107. In our project we are using an IC in place of bridge rectifier. The picture of DB 107 is shown in fig: 3.7

Features:

- Good for automation insertion
- Surge overload rating - 30 amperes peak
- Ideal for printed circuit board
- Reliable low cost construction utilizing molded
- Glass passivated device
- Polarity symbols molded on body
- Mounting position: Any
- Weight: 1.0 gram

Fig 3.7: DB107

Step 3: Filtration

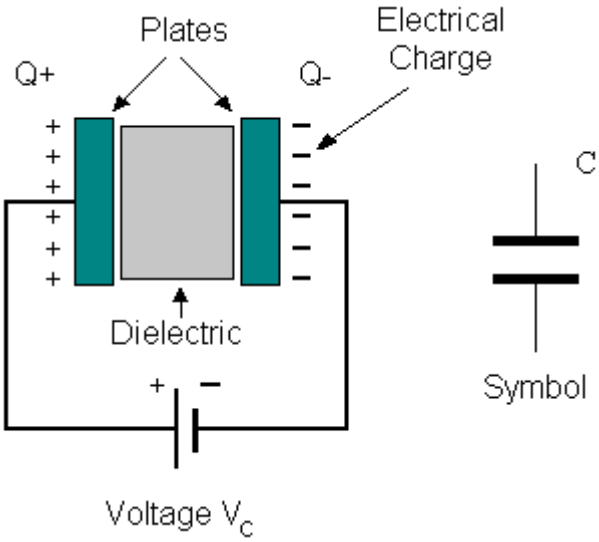
The process of converting a pulsating direct current to a pure direct current using filters is called as filtration.

Filters:

Electronic filters are electronic circuits, which perform signal-processing functions, specifically to remove unwanted frequency components from the signal, to enhance wanted ones.

Introduction to Capacitors:

The Capacitor or sometimes referred to as a Condenser is a passive device, and one which stores energy in the form of an electrostatic field which produces a potential (static voltage) across its plates. In its basic form a capacitor consists of two parallel conductive plates that are not connected but are electrically separated either by air or by an insulating material called the Dielectric. When a voltage is applied to these plates, a current flows charging up the plates with electrons giving one plate a positive charge and the other plate an equal and opposite negative charge this flow of electrons to the plates is known as the Charging Current and continues to flow until the voltage across the plates (and hence the capacitor) is equal to the applied voltage Vcc. At this point the capacitor is said to be fully charged and this is illustrated below. The construction of capacitor and an electrolytic capacitor are shown in figures 3.8 and 3.9 respectively.



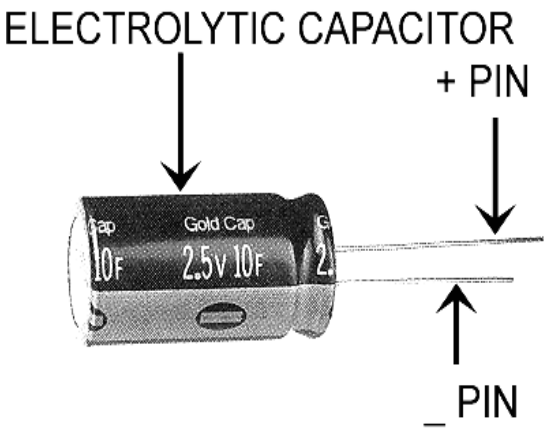


Fig 3.8:Construction Of a Capacitor **Fig 3.9:Electrolytic Capaticor**

Step 4: Regulation

The process of converting a varying voltage to a constant regulated voltage is called as regulation. For the process of regulation we use voltage regulators.

Voltage Regulator:

A voltage regulator (also called a 'regulator') with only three terminals appears to be a simple device, but it is in fact a very complex integrated circuit. It converts a varying input voltage into a constant 'regulated' output voltage. Voltage Regulators are available in a variety of outputs like 5V, 6V, 9V, 12V and 15V. The LM78XX series of voltage regulators are designed for positive input. For applications requiring negative input, the LM79XX series is used. Using a pair of 'voltage-divider' resistors can increase the output voltage of a regulator circuit.

It is not possible to obtain a voltage lower than the stated rating. You cannot use a 12V regulator to make a 5V power supply. Voltage regulators are very robust. These can withstand over-current draw due to short circuits and also over-heating. In both cases, the regulator will cut off before any damage occurs. The only way to destroy a regulator is to apply reverse voltage to its input. Reverse polarity destroys the regulator almost instantly. Fig: 3.10 shows voltage regulator.

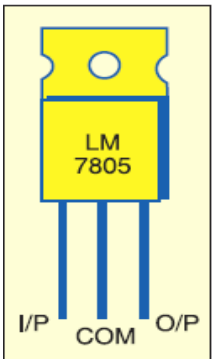


Fig 3.10: Voltage Regulator

Resistors:

A resistor is a two-terminal electronic component that produces a voltage across its terminals that is proportional to the electric current passing through it in accordance with Ohm's law:

$$V = IR$$

Resistors are elements of electrical networks and electronic circuits and are ubiquitous in most electronic equipment. Practical resistors can be made of various compounds and films, as well as resistance wire (wire made of a high-resistivity alloy, such as nickel/chrome).

The primary characteristics of a resistor are the resistance, the tolerance, maximum working voltage and the power rating. Other characteristics include temperature coefficient, noise, and inductance. Less well-known is critical resistance, the value below which power dissipation limits the maximum permitted current flow, and above which the limit is applied voltage. Critical resistance is determined by the design, materials and dimensions of the resistor.

3.4. LED:

A light-emitting diode (LED) is a semiconductor light source. LEDs are used as indicator lamps in many devices, and are increasingly used for lighting. Introduced as a practical electronic component in 1962, early LEDs emitted low-intensity red light, but modern versions are available across the visible, ultraviolet and infrared wavelengths, with very high brightness. The internal structure and parts of a led are shown below.

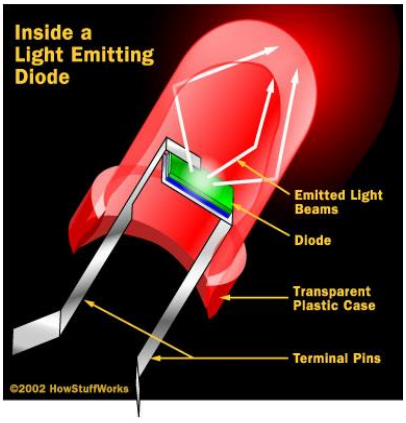


Fig 3.11: Inside a LED

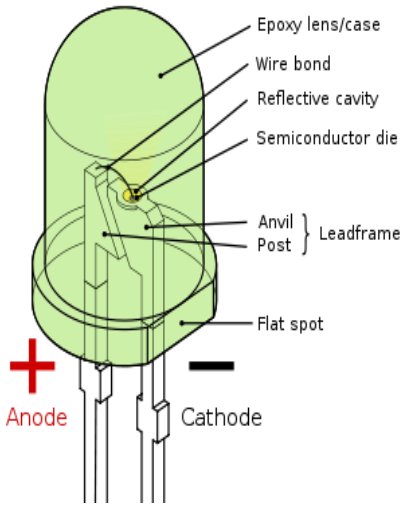


Fig 3.12: Parts of a LED

Applications of LED fall into three major categories:

- Visual signal application where the light goes more or less directly from the LED to the human eye, to convey a message or meaning.
- Illumination where LED light is reflected from object to give visual response of these objects.
- Generate light for measuring and interacting with processes that do not involve the human visual system.

3.5: Potentiometer

A **potentiometer** informally a **pot** is a three-terminal resistor with a sliding contact that forms an adjustable voltage divider. If only two terminals are used, one end and the wiper, it acts as a **variable resistor** or **rheostat**.

Potentiometers are commonly used to control electrical devices such as volume controls on audio equipment. Potentiometers operated by a mechanism can be used as position transducers, for example, in a joystick.

A potentiometer measuring instrument is essentially a voltage divider used for measuring electric potential (voltage); the component is an implementation of the same principle, hence its name.

Fig 3.13:POT(Potentiometer)

Potentiometers are commonly used to control electrical devices such as volume controls on audio equipment. Potentiometers operated by a mechanism can be used as position transducers, for example, in a joystick. Potentiometers are rarely used to directly control significant power

(more than a watt), since the power dissipated in the potentiometer would be comparable to the power in the controlled load.

Features:

- 10K ohm linear taper potentiometer.
- Provides level adjustment of voltage controlled amplifiers from a single Remote location.
- Continuous, rotary control with stops at 0 and Max plus incremental detents for precision feel.

Potentiometer applications:

Potentiometers are rarely used to directly control significant amounts of power (more than a watt or so). Instead they are used to adjust the level of analog signals (for example volume controls on audio equipment), and as control inputs for electronic circuits. For example, a light dimmer uses a potentiometer to control the switching of a TRIAC and so indirectly to control the brightness of lamps.

Preset potentiometers are widely used throughout electronics wherever adjustments must be made during manufacturing or servicing.

User-actuated potentiometers are widely used as user controls, and may control a very wide variety of equipment functions. The widespread use of potentiometers in consumer electronics declined in the 1990s, with rotary encoders, up/down push-buttons, and other digital controls now more common. However they remain in many applications, such as volume controls and as position sensors.

1. Audio control:

Fig 3.14 : Linear potentiometers ("faders")

Low-power potentiometers, both linear and rotary, are used to control audio equipment, changing loudness, frequency attenuation and other characteristics of audio signals.

The 'log pot' is used as the volume control in audio power amplifiers, where it is also called an "audio taper pot", because the amplitude response of the human ear is approximately logarithmic. It ensures that on a volume control marked 0 to 10, for example, a setting of 5 sounds subjectively half as loud as a setting of 10. There is also an anti-log pot or reverse audio taper

which is simply the reverse of a logarithmic potentiometer. It is almost always used in a ganged configuration with a logarithmic potentiometer, for instance, in an audio balance control.

Potentiometers used in combination with filter networks act as tone controls or equalizers.

2. Television:

Potentiometers were formerly used to control picture brightness, contrast, and color response. A potentiometer was often used to adjust "vertical hold", which affected the synchronization between the receiver's internal sweep circuit (sometimes a multi vibrator) and the received picture signal, along with other things such as audio-video carrier offset, tuning frequency (for push-button sets) and so on.

3. Motion Control:

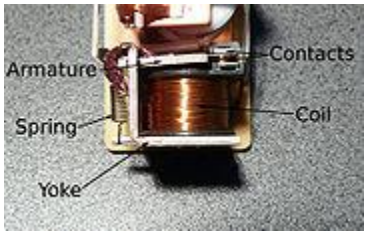
Potentiometers can be used as position feedback devices in order to create "closed loop" control, such as in a servomechanism.

3.6 Relay:

A **relay** is an electrically operated switch. Many relays use an electromagnet to operate a switching mechanism, but other operating principles are also used. Relays find applications where it is necessary to control a circuit by a low-power signal, or where several circuits must be controlled by one signal. The first relays were used in long distance telegraph circuits, repeating the signal coming in from one circuit and re-transmitting it to another. Relays found extensive use in telephone exchanges and early computers to perform logical operations. A type of relay that can handle the high power required to directly drive an electric motor is called a contactor. Solid-state relays control power circuits with no moving parts, instead using a semiconductor device triggered by light to perform switching. Relays with calibrated operating characteristics and sometimes multiple operating coils are used to protect electrical circuits from overload or faults; in modern electric power systems these functions are performed by digital instruments still called "protection relays".

3.6.1 Basic design and operation:

1. **Simple electromechanical relay**



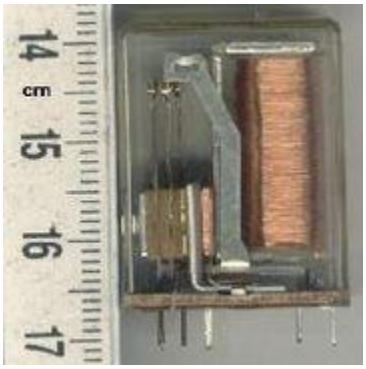


Fig 3.15:Simple Electromechanical Relay

A simple electromagnetic relay, such as the one taken from a car in the first picture, is an adaptation of an electromagnet. It consists of a coil of wire surrounding a soft iron core, an iron yoke, which provides a low reluctance path for magnetic flux, a movable iron armature, and a set, or sets, of contacts; two in the relay pictured. The armature is hinged to the yoke and mechanically linked to a moving contact or contacts. It is held in place by a spring so that when the relay is de-energized there is an air gap in the magnetic circuit. In this condition, one of the two sets of contacts in the relay pictured is closed, and the other set is open. Other relays may have more or fewer sets of contacts depending on their function. The relay in the picture also has a wire connecting the armature to the yoke. This ensures continuity of the circuit between the moving contacts on the armature, and the circuit track on the printed circuit board (PCB) via the yoke, which is soldered to the PCB.

3.6.2 Types of Relays:

1. **Latching relay**

Fig 3.16:Latching Relay

A **latching relay** has two relaxed states (bi stable). These are also called "impulse", "keep", or "stay" relays. When the current is switched off, the relay remains in its last state. This is achieved with a solenoid operating a ratchet and cam mechanism, or by having two opposing coils with an over-center spring or permanent magnet to hold the armature and contacts in position while the coil is relaxed, or with a remnant core. In the ratchet and cam example, the first pulse to the coil turns the relay on and the second pulse turns it off. In the two coil example, a pulse to one coil turns the relay on and a pulse to the opposite coil turns the relay off. This type of relay has the advantage that it consumes power only for an instant, while it is being switched, and it retains its last setting across a power outage. A remnant core latching relay requires a current pulse of opposite polarity to make it change state.

2. Contactor relay

A **contactor** is a very heavy-duty relay used for switching electric motors and lighting loads. Continuous current ratings for common contactors range from 10 amps to several hundred amps. High-current contacts are made with alloys containing silver. The unavoidable arcing causes the contacts to oxidize; however, silver oxide is still a good conductor. Such devices are often used for motor starters. A motor starter is a contactor with overload protection devices attached. The overload sensing devices are a form of heat operated relay where a coil heats a bi-metal strip, or where a solder pot melts, releasing a spring to operate auxiliary contacts. These auxiliary contacts are in series with the coil. If the overload senses excess current in the load, the coil is de-

energized. Contactor relays can be extremely loud to operate, making them unfit for use where noise is a chief concern.

3. Solid-state relay

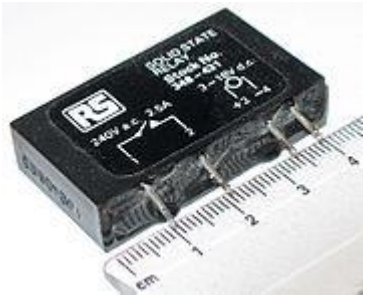


Fig 3.17:Solid State Relay

Solid state relay, which has no moving parts

25 A or 40 A solid state contactors

A **solid state relay** (**SSR**) is a solid state electronic component that provides a similar function to an electromechanical relay but does not have any moving components, increasing long-term reliability. With early SSR's, the tradeoff came from the fact that every transistor has a small voltage drop across it. This voltage drop limited the amount of current a given SSR could handle. As transistors improved, higher current SSR's, able to handle 100 to 1,200 Amperes, have become commercially available. Compared to electromagnetic relays, they may be falsely triggered by transients.

4. Buchholz relay

A **Buchholz relay** is a safety device sensing the accumulation of gas in large oil-filled transformers; this will alarm on slow accumulation of gas or shut down the transformer if gas is produced rapidly in the transformer oil.

5. Overload protection relay

Electric motors need over current protection to prevent damage from over-loading the motor, or to protect against short circuits in connecting cables or internal faults in the motor windings. One type of electric motor overload protection relay is operated by a heating element in series with the electric motor. The heat generated by the motor current heats a bimetallic strip or melts solder, releasing a spring to operate contacts. Where the overload relay is exposed to the same environment as the motor, a useful though crude compensation for motor ambient temperature is provided.

3.6.3 Applications of Relays:

- Control a high-voltage circuit with a low-voltage signal, as in some types of modems or audio amplifiers,
- Control a high-current circuit with a low-current signal, as in the starter solenoid of an automobile,
- Detect and isolate faults on transmission and distribution lines by opening and closing circuit breakers (protection relays),

Advantages of relays:

- Relays can switch **AC and DC**, transistors can only switch DC.
- Relays can switch **high voltages**, transistors cannot.
- Relays are a better choice for switching **large currents** (> 5A).
- Relays can switch **many contacts** at once.

Disadvantages of relays:

- Relays are **bulkier** than transistors for switching small currents.

- Relays **cannot switch rapidly** (except reed relays), transistors can switch many times per second.
- Relays **use more power** due to the current flowing through their coil.
- Relays **require more current than many ICs can provide**, so a low power transistor may be needed to switch the current for the relay's coil.

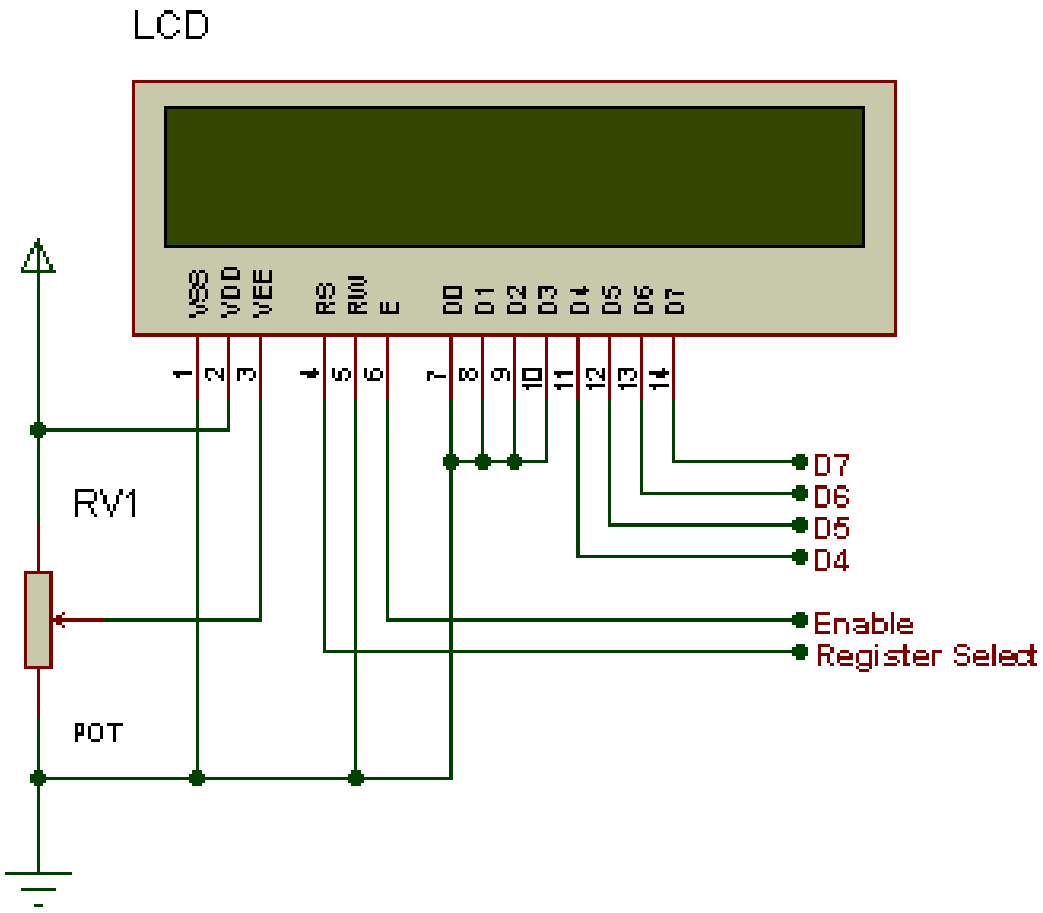
3.7 LCD DISPLAY

LCD Background:

One of the most common devices attached to a micro controller is an LCD display. Some of the most common LCD's connected to the many microcontrollers are 16x2 and 20x2 displays. This means 16 characters per line by 2 lines and 20 characters per line by 2 lines, respectively.

Basic 16x 2 Characters LCD

Fig 3.18: LCD Pin diagram



Pin description:

Pin No.	Name	Description
Pin no. 1	**VSS**	Power supply (GND)
Pin no. 2	**VCC**	Power supply (+5V)
Pin no. 3	**VEE**	Contrast adjust
Pin no. 4	**RS**	0 = Instruction input 1 = Data input
Pin no. 5	**R/W**	0 = Write to LCD module 1 = Read from LCD module
Pin no. 6	**EN**	Enable signal
Pin no. 7	**D0**	Data bus line 0 (LSB)
Pin no. 8	**D1**	Data bus line 1
Pin no. 9	**D2**	Data bus line 2
Pin no. 10	**D3**	Data bus line 3
Pin no. 11	**D4**	Data bus line 4
Pin no. 12	**D5**	Data bus line 5
Pin no. 13	**D6**	Data bus line 6
Pin no. 14	**D7**	Data bus line 7 (MSB)

3.8 WI-FI Module - 8266

The ESP8266 Wi-Fi Module is a self contained SOC with integrated TCP/IP protocol stack that can give any microcontroller access to your Wi-Fi network. The ESP8266 is capable of either hosting an application or offloading all Wi-Fi networking functions from another application processor.

Each ESP8266 module comes pre-programmed with an AT command set firmware.

Features

- 802.11 b/g/n
- Wi-Fi Direct (P2P), soft-AP
- Integrated TCP/IP protocol stack

- Integrated PLLs, regulators, DCXO and power management units
- +19.5dBm output power in 802.11b mode
- Power down leakage current of <10uA
- 1MB Flash Memory

3.9 Thingspeak

ThingSpeak is an Internet of Things (IoT) platform that allows users to collect, analyze, and visualize data from sensors and other devices in real-time. It provides an easy-to-use interface for creating IoT applications and has gained popularity for its versatility and integration capabilities. With features like data logging, MATLAB analytics, and customizable visualizations, ThingSpeak empowers individuals and organizations to harness the power of their data to make informed decisions and develop innovative solutions.

CHAPTER 4: SOFTWARE DESCRIPTION

This project is implemented using following software's:

- Express PCB – for designing circuit
- PIC C compiler - for compilation part

4.1 The Interface:

When a project is first started you will be greeted with a yellow outline. This yellow outline is the dimension of the PCB. Typically after positioning of parts and traces, move them to their final position and then crop the PCB to the correct size. However, in designing a board with a certain size constraint, crop the PCB to the correct size before starting.

Fig: 4.1 show the toolbar in which the each button has the following functions:

Fig 4.1: Tool bar necessary for the interface

- The select tool: It is fairly obvious what this does. It allows you to move and manipulate parts. When this tool is selected the top toolbar will show buttons to move traces to the top / bottom copper layer, and rotate buttons.
- The zoom to selection tool: does just that.
- The place pad: button allows you to place small soldier pads which are useful for board connections or if a part is not in the part library but the part dimensions are available. When this tool is selected the top toolbar will give you a large selection of round holes, square holes and surface mount pads.
- The place component: tool allows you to select a component from the top toolbar and then by clicking in the workspace places that component in the orientation chosen using the buttons next to the component list. The components can always be rotated afterwards with the select tool if the orientation is wrong.
- The place trace: tool allows you to place a solid trace on the board of varying thicknesses. The top toolbar allows you to select the top or bottom layer to place the trace on.

- The Insert Corner in trace: button does exactly what it says. When this tool is selected, clicking on a trace will insert a corner which can be moved to route around components and other traces.
- The remove a trace button is not very important since the delete key will achieve the same result.

4.2 PIC Compiler:

PIC compiler is software used where the machine language code is written and compiled. After compilation, the machine source code is converted into hex code which is to be dumped into the microcontroller for further processing. PIC compiler also supports C language code.

It's important that you know C language for microcontroller which is commonly known as Embedded C. As we are going to use PIC Compiler, hence we also call it PIC C. The PCB, PCM, and PCH are separate compilers. PCB is for 12-bit opcodes, PCM is for 14-bitopcodes, and PCH is for 16-bit opcode PIC microcontrollers. Due to many similarities, all three compilers are covered in this reference manual. Features and limitations that apply to only specific microcontrollers are indicated within. These compilers are specifically designed to meet the unique needs of the PIC microcontroller. This allows developers to quickly design applications software in a more readable, high-level language. When compared to a more traditional C compiler, PCB, PCM, and PCH have some limitations. As an example of the limitations, function recursion is not allowed.

This is due to the fact that the PIC has no stack to push variables onto, and also because of the way the compilers optimize the code. The compilers can efficiently implement normal C constructs, input/output operations, and bit twiddling operations. All normal C data types are supported along with pointers to constant arrays, fixed point decimal, and arrays of bits.

PIC C is not much different from a normal C program. If you know assembly, writing a C program is not a crisis. In PIC, we will have a main function, in which all your application specific work will be defined. In case of embedded C, you do not have any operating system running in there. So you have to make sure that your program or main file should never exit. This can be done with the help of simple while (1) or for loop as they are going to run infinitely.

We have to add header file for controller you are using, otherwise you will not be able to access registers related to peripherals.

#include <16F73.h> // header file for PIC 16F73//

4.3 Procedural steps for compilation, simulation and dumping:

4.3.1 Compilation and simulation steps:

For PIC microcontroller, PIC C compiler is used for compilation. The compilation steps are as follows:

- Open PIC C compiler.
- You will be prompted to choose a name for the new project, so create a separate folder where all the files of your project will be stored, choose a name and click save.

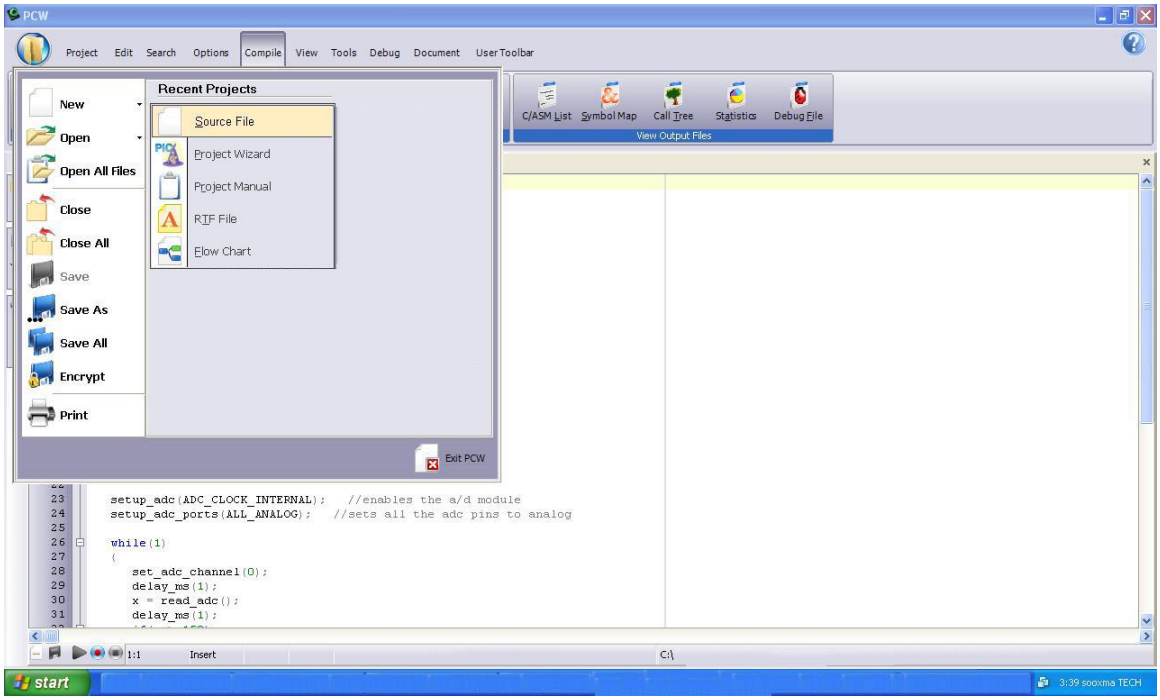


Fig 4. 2: Picture of opening a new file using PIC C compiler

- Click Project**,** New, and something the box named 'Text1' is where your code should be written later.
- Now you have to click 'File, Save as' and choose a file name for your source code ending with the letter '.c'. You can name as 'project.c' for example and click save. Then you have to add this file to your project work.

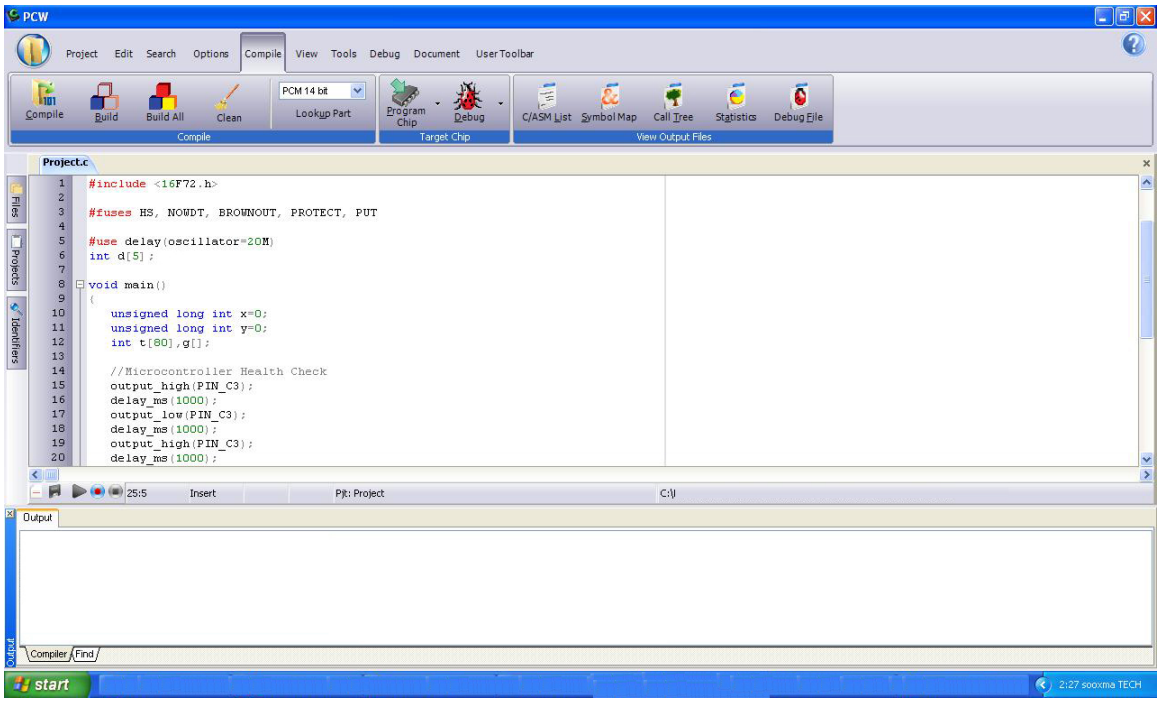


Fig 4.3: Picture of compiling a new file using PIC C compiler

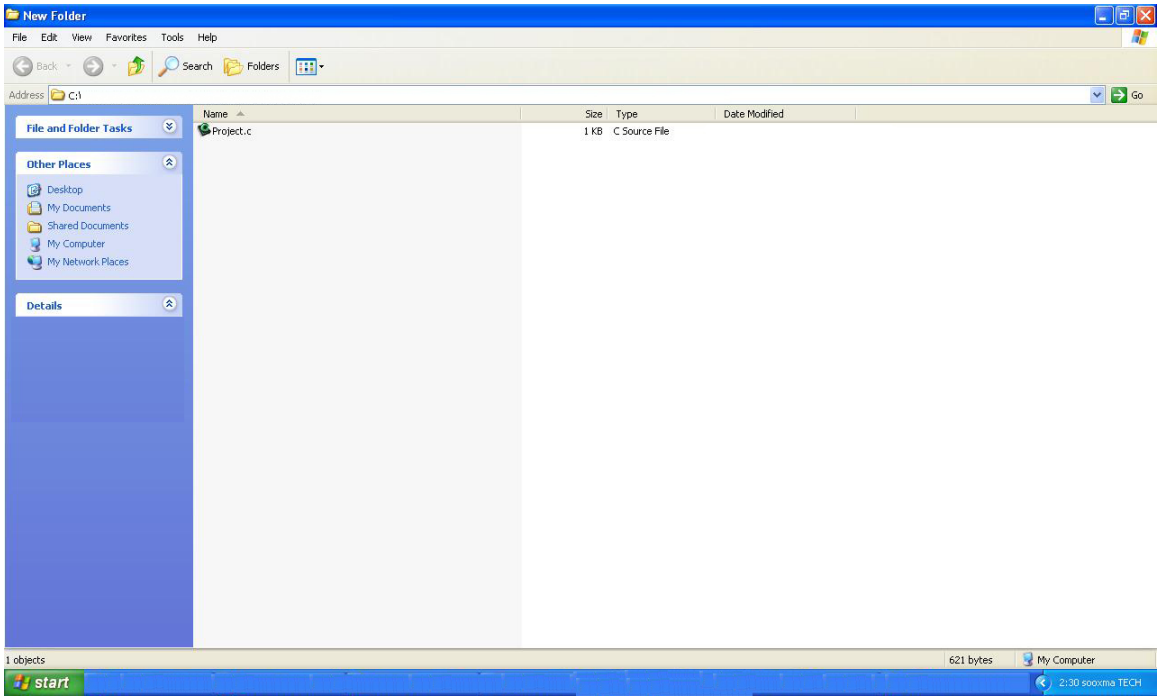


Fig 4.4: Picture of compiling a project.c file using PIC C compiler

- You can then start to write the source code in the window titled 'project.c' then before testing your source code; you have to compile your source code, and correct eventual syntax errors.

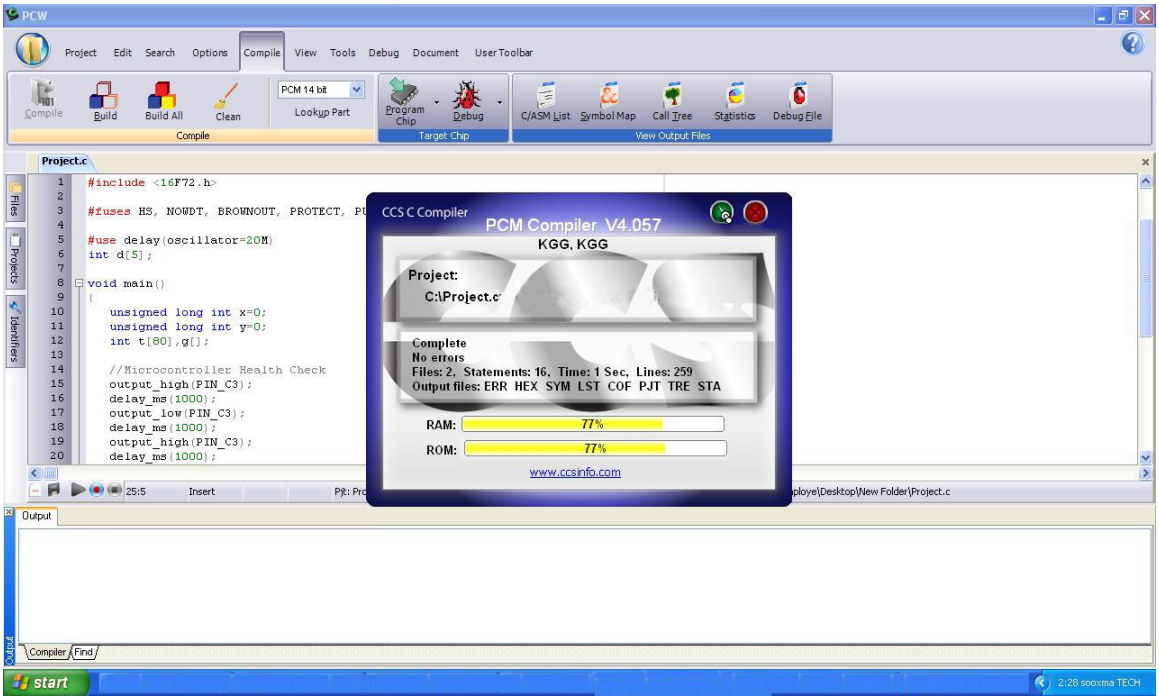


Fig 4.5: Picture of checking errors and warnings using PIC C compiler

- By clicking on compile option .hex file is generated automatically.
- This is how we compile a program for checking errors and hence the compiled program is saved in the file where we initiated the program.

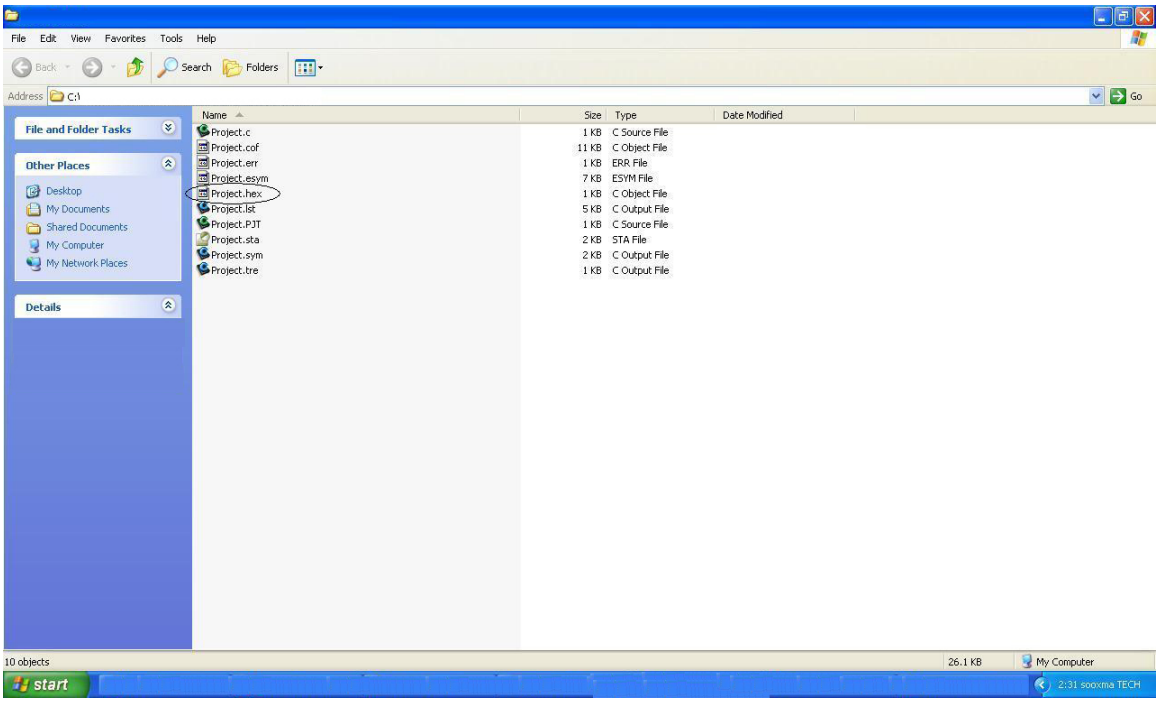


Fig 4.6: Picture of .hex file existing using PIC C compiler

After compilation, next step is simulation. Here first circuit is designed in Express PCB using Proteus 7 software and then simulation takes place followed by dumping. The simulation steps are as follows:

- Open Proteus 7 and click on IS1S6.
- Now it displays PCB where circuit is designed using microcontroller. To design circuit components are required. So click on component option.

10. Now click on letter 'p', then under that select PIC16F73 ,other components related to the project and click OK. The PIC 16F73 will be called your "Target device", which is the final destination of your source code.

4.3.2 Dumping steps:

The steps involved in dumping the program edited in proteus 7 to microcontroller are shown below:

1. Initially before connecting the program dumper to the microcontroller kit the window is appeared as shown below.

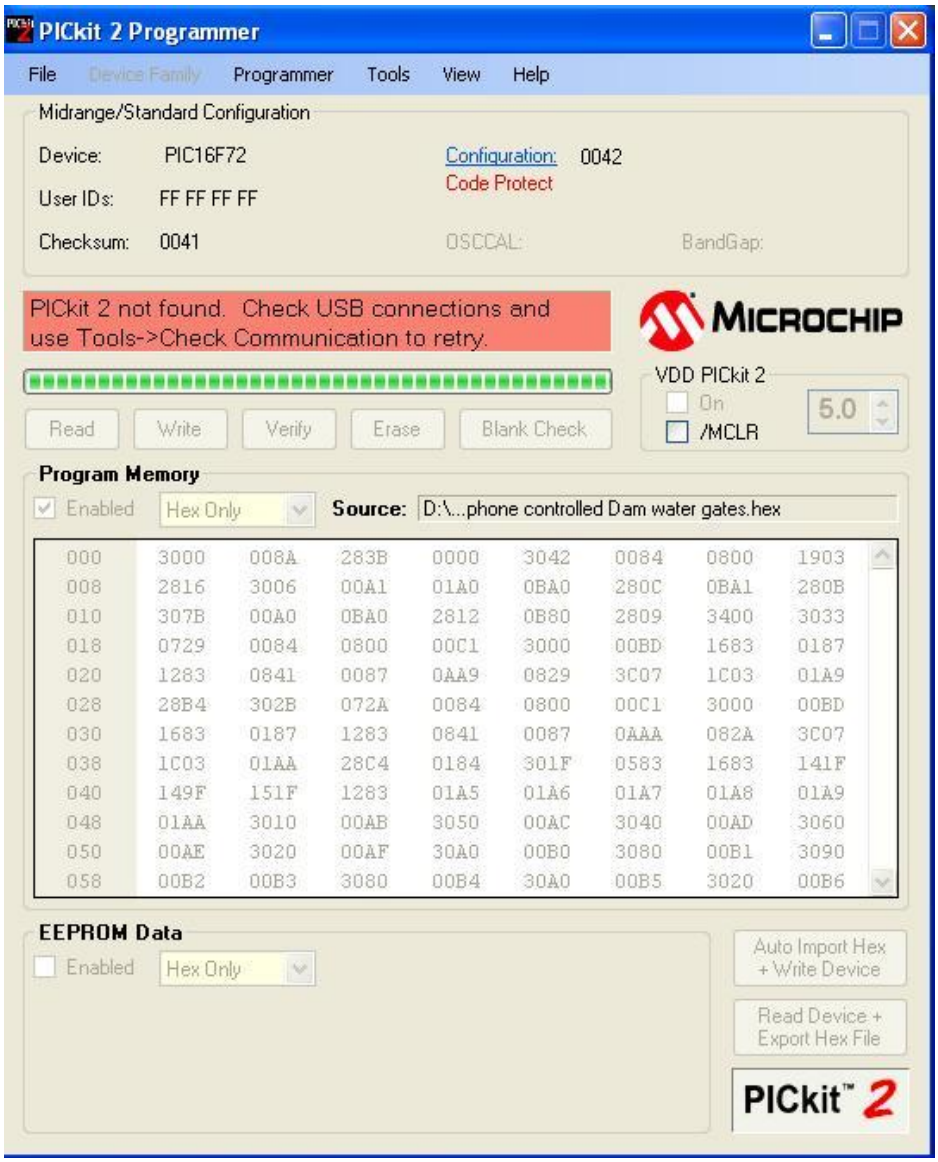


Fig 4.7: Picture of program dumper window

2 .Select Tools option and click on Check Communication for establishing a connection as shown in below window

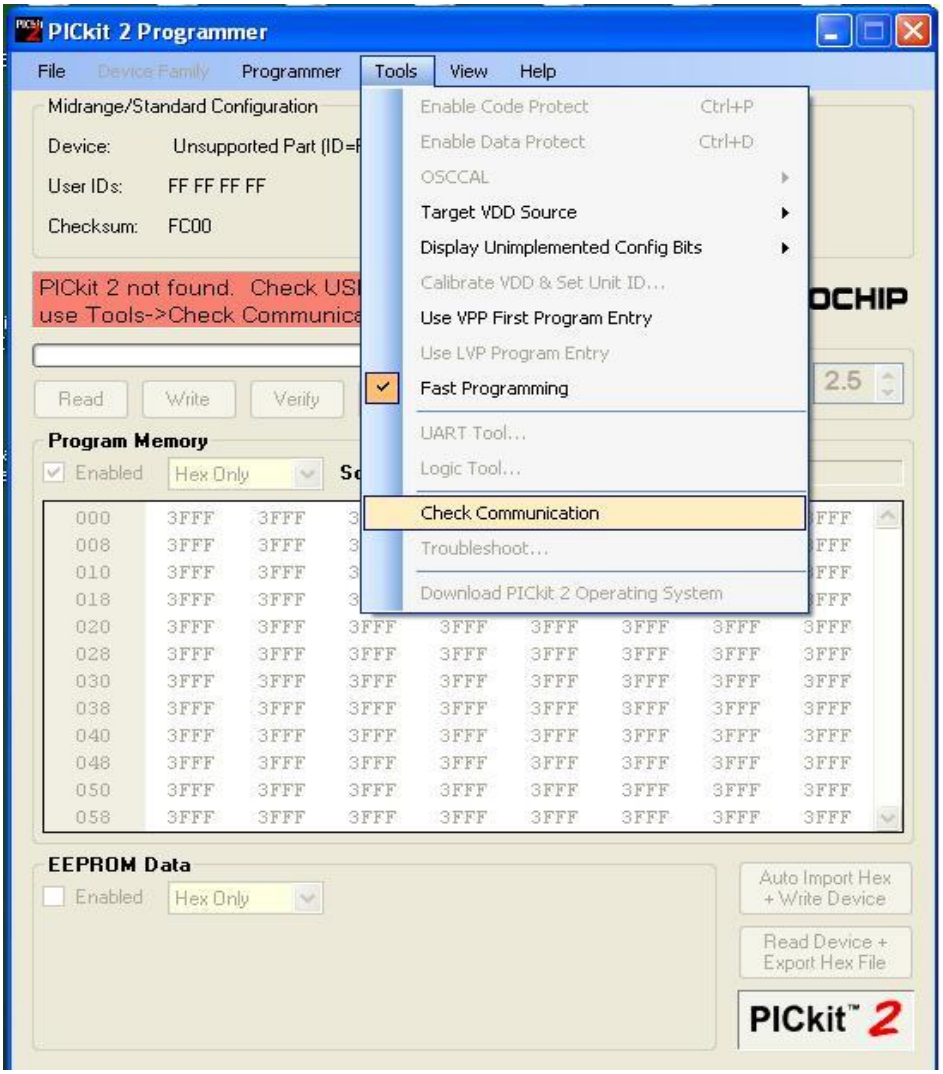


Fig 4.8: Picture of checking communications before dumping program into microcontroller

3. After connecting the dumper properly to the microcontroller kit the window is appeared as shown below.

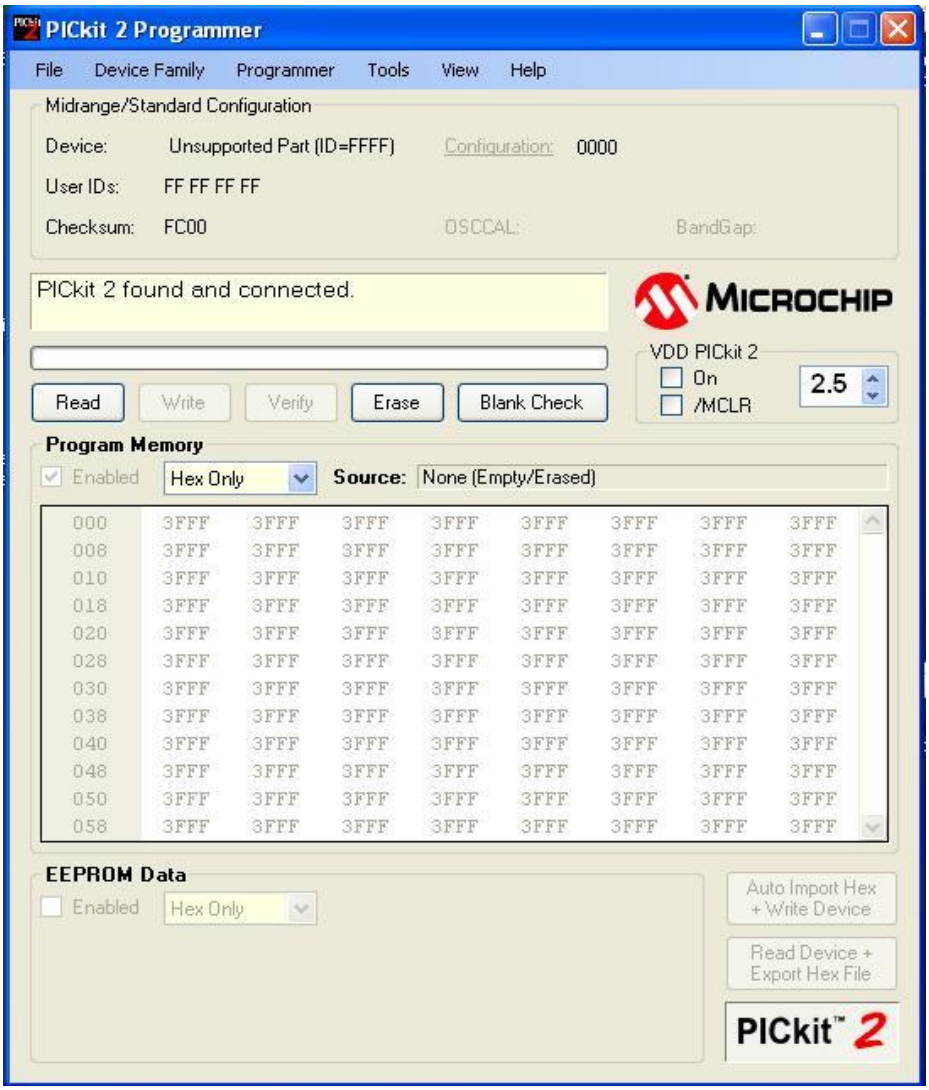


Fig 4.9: Picture after connecting the dumper to microcontroller

4. Again by selecting the Tools option and clicking on Check Communication the microcontroller gets recognized by the dumper and hence the window is as shown below.

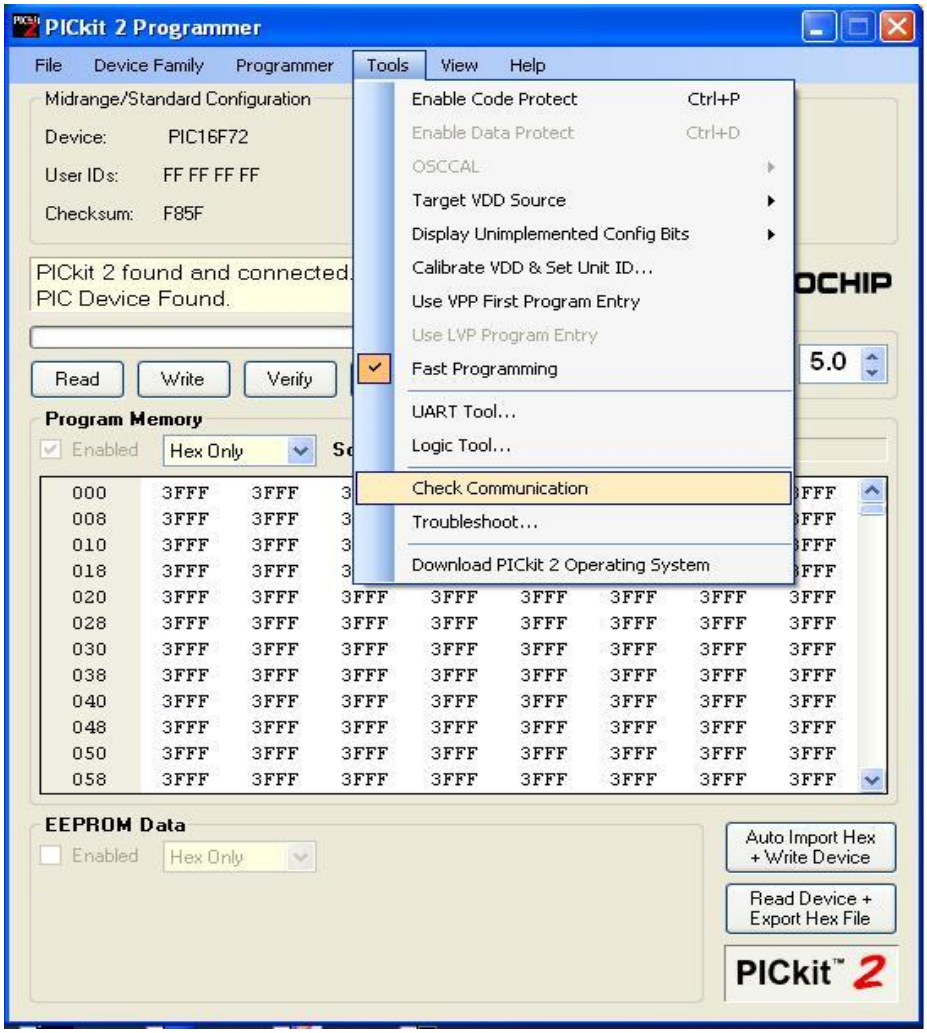


Fig 4.10: Picture of dumper recognition to microcontroller

5. Import the program which is '.hex' file from the saved location by selecting File option and clicking on 'Import Hex' as shown in below window.

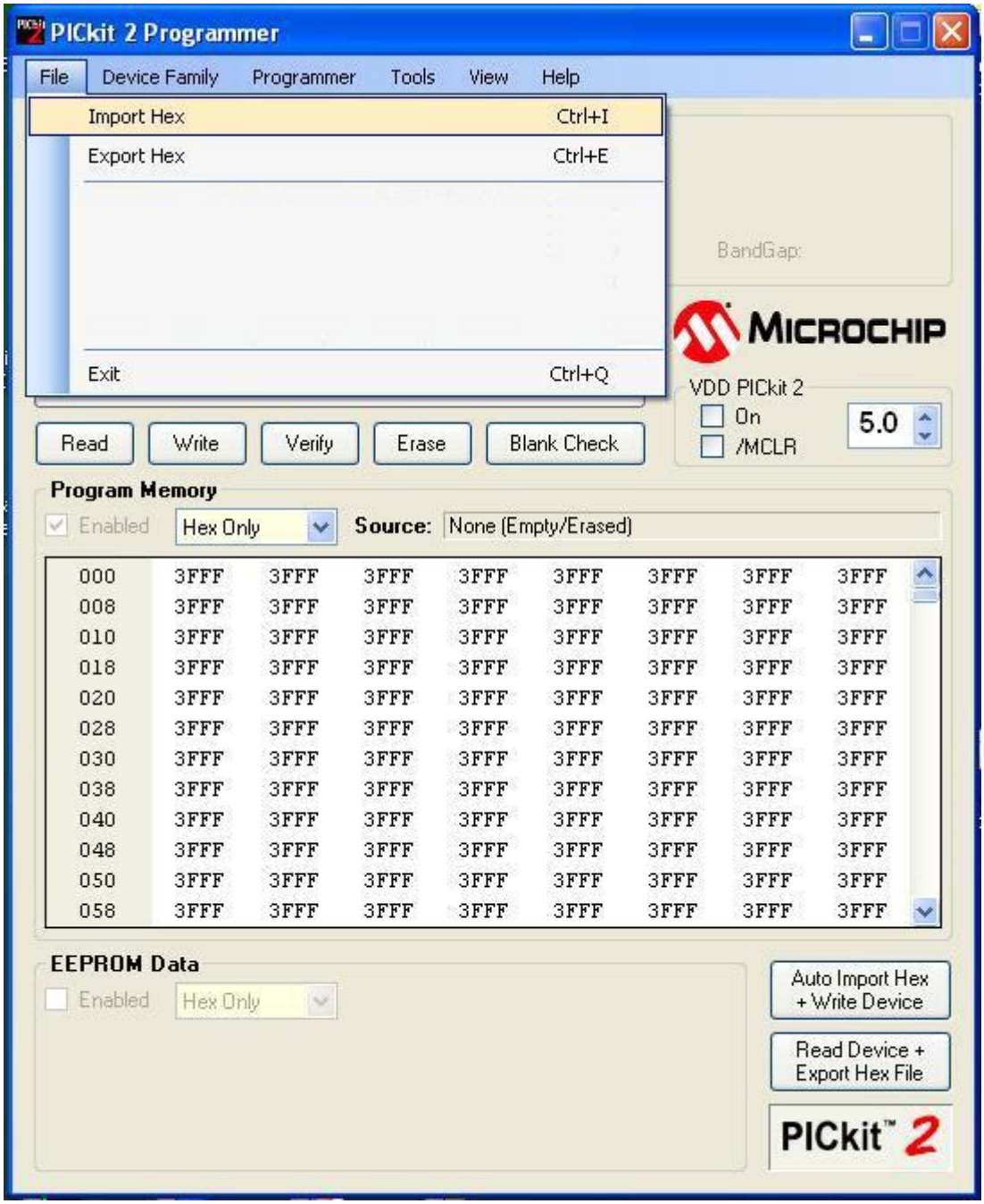


Fig 4.11: Picture of program importing into the microcontroller

6. After clicking on 'Import Hex' option we need to browse the location of our program and click the 'prog.hex' and click on 'open' for dumping the program into the microcontroller.

7. After the successful dumping of program the window is as shown below.

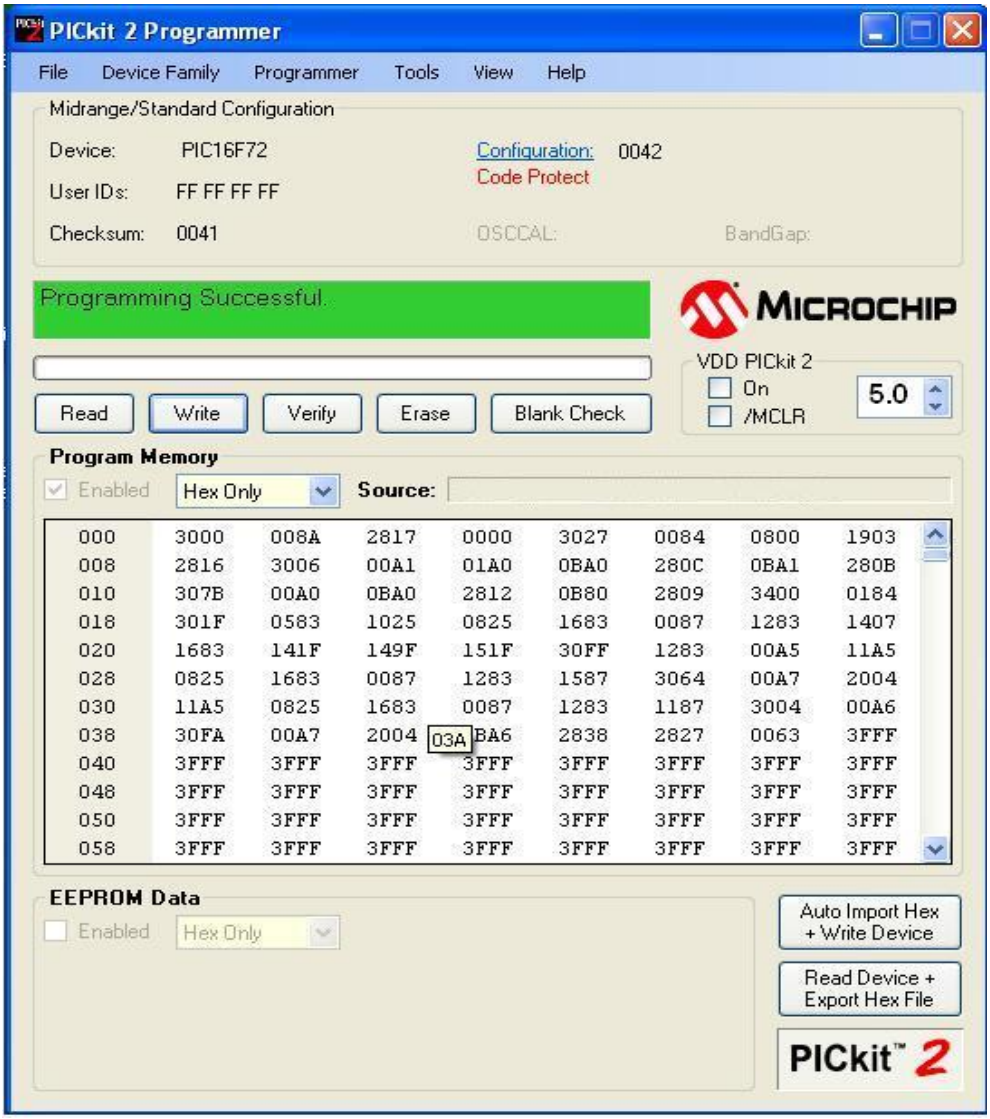


Fig 4.12: Picture after program dumped into the microcontroller

CHAPTER 5: PROJECT DESCRIPTION

In this chapter, schematic diagram and interfacing of PIC16F73 microcontroller with each module is considered**.**

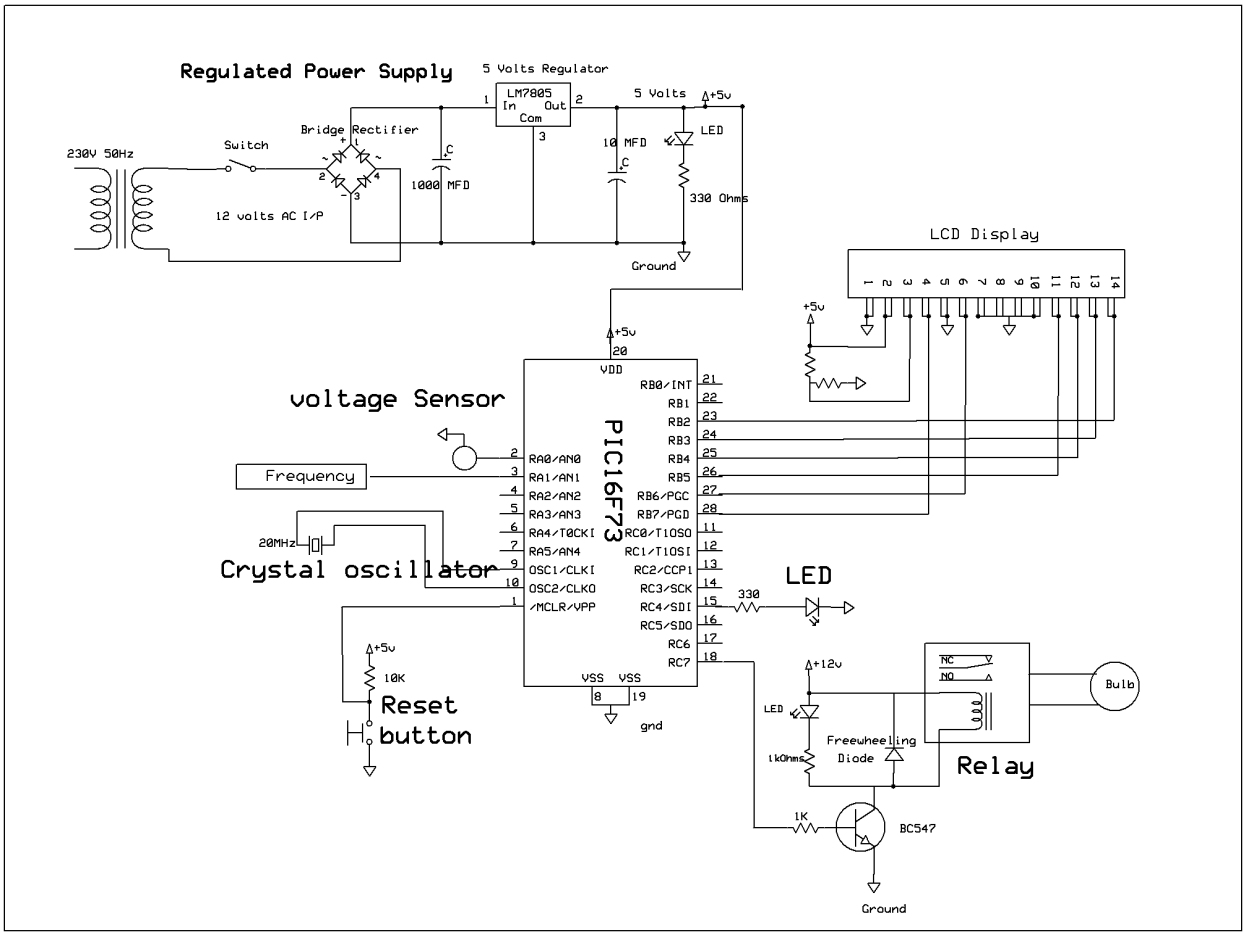


Fig 5.1: Schematic diagram

The above schematic diagram of **Power Grid Failure detection based on sensing frequency and voltage variation using IoT** explains the interfacing section of each component with micro controller and RF module. The crystal oscillator connected to 9th and 10th pins of micro controller and regulated power supply is also connected to micro controller and LEDs also connected to micro controller through resistors and motor driver connected to micro controller

Here regulated power supply (RPS) is used which is block of a transformer, rectifier, filter and regulator The 3-phase, 220V supply is converted to 5V using step down transformer and it is given to the bridge rectifier for conversion of 220V AC to 5V DC and filter is connected parallel to remove unwanted ripples and harmonics this 5V DC is given to the PIC microcontroller because it is designed to operate in 5V range. The neutral is commonly connected to the bulb and RPS, the phase is given to the bulb by connecting a relay parallel to the RPS.

The frequency port is connected to the comparator board to compare and generate the frequency and voltage port is connected to the PIC microcontroller, one can arise the doubt that why both the ports is connected to the PIC microcontroller because it can compare only one variable at a time. The predetermined values of voltage and frequency are set to the PIC microcontroller if voltage or frequency level goes beyond this predetermined ranges the PIC microcontroller turns off the bulb using relay by connecting PIC microcontroller to the relay.

The voltage and frequency levels are shown in the LCD display it also mentions if voltage or frequency levels are in abnormal conditions whether it is over/under voltage and high/low frequency. A wifi module is connected to the PIC microcontroller such that voltage and frequency graphs are shown in the smart phone using IoT we can see this graphs in ThingSpeak it is an Internet of Things (IoT) platform that allows users to collect, analyze, and visualize data from sensors and other devices in real-time.

CHAPTER 6: ADVANTAGES AND DISADVANTAGES

Advantages:

- Devices can be operated using relay.
- Monitoring parameters like voltage, frequency etc.
- Visual alerts through LCD display unit.
- Efficient and low-cost design.
- Low power consumption.
- Real time monitoring and controlling.

Disadvantages:

1. Interfacing sensors to microcontroller is sensitive

Applications:

1. This system can be implemented in industries.
2. This system can be used to monitoring and controlling the home appliances.

CHAPTER 7: RESULTS

7.1 Result:

The project "**Power Grid Failure detection based on sensing frequency and voltage variation using IoT**" was designed such that the devices can be monitored and also controlled various electrical devices within the substation and the status of the devices are displayed on LCD.

Voltage detection is by done by varying the potentiometer after reaching the acceptable range the LCD displays that the voltage is EXCEED 230V and the relay will be tripped and load of AC is protected. The frequency detection is done before the tripping of the light load the light flicker and frequency change will be displayed on the LCD. Hence a continuous monitoring load and faults in frequency and voltage is done by using microcontroller. In this case the load variation will be sensed. For varying the load in this prototype we are using LED's and DC lamp load.

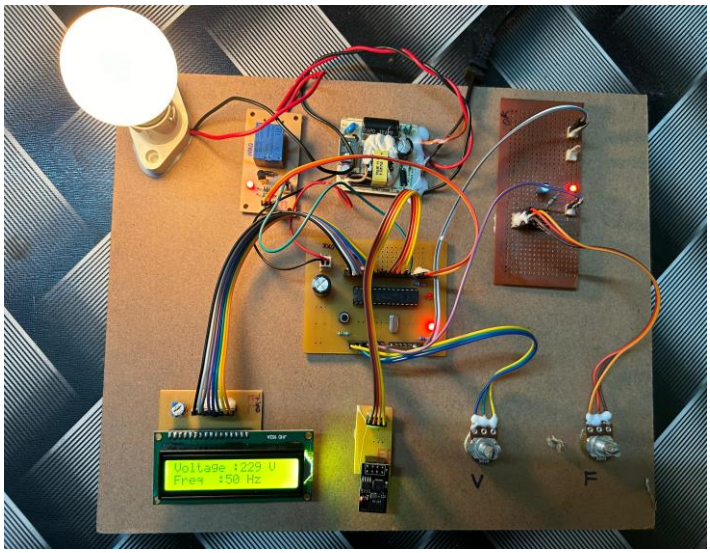


Fig.7.1: Hard ware model at normal condition

Our project calls for a frequency of 49Hz-51Hz and a voltage range of 210V–245V in order to synchronize a producing station with the grid. When the input voltage drops below 210V or rises beyond 245V, the load is turned off immediately. The load is immediately cut off if the frequency drops below 49Hz or above 51Hz.

As shown in the Fig.7.1 The level of voltage is 229V and the frequency is 50Hz, those are at synchronous condition, so that the bulb is ON.

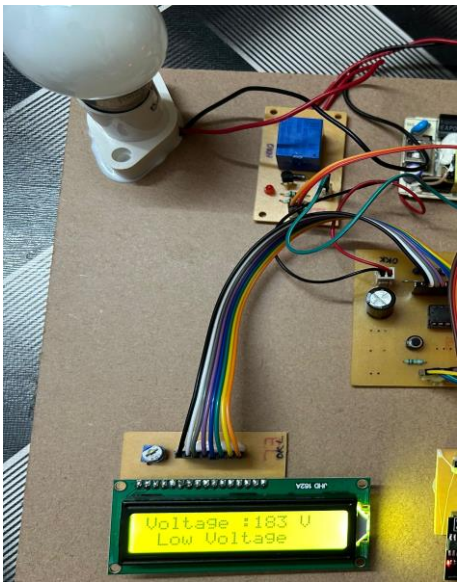


Fig 7.2: Circuit at Low Voltage

As shown in the Fig.7.2 The level of voltage is 183V and the frequency is 50Hz, those are at Asynchronous condition, so that the bulb is OFF.

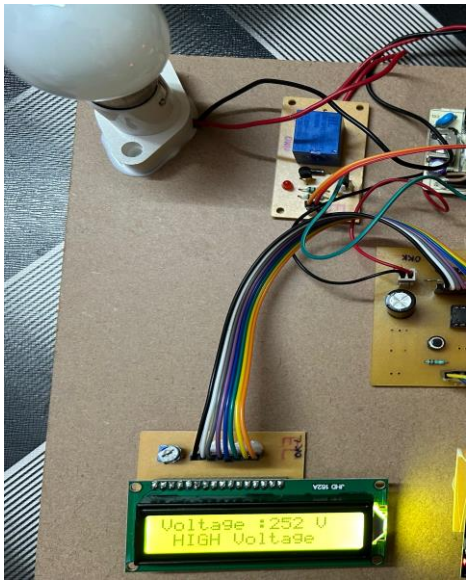
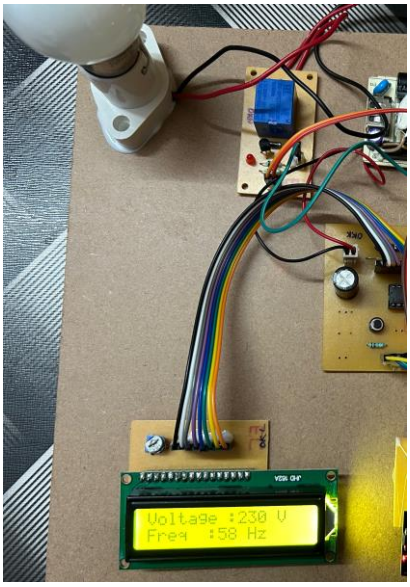


Fig 7.3: Circuit at High Voltage

As shown in the Fig.7.3 The level of voltage is 252v and the frequency is 50 Hz, those are at Asynchronous condition, so that the bulb is OFF.



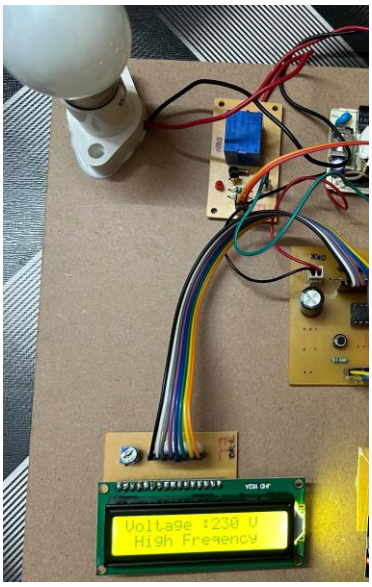
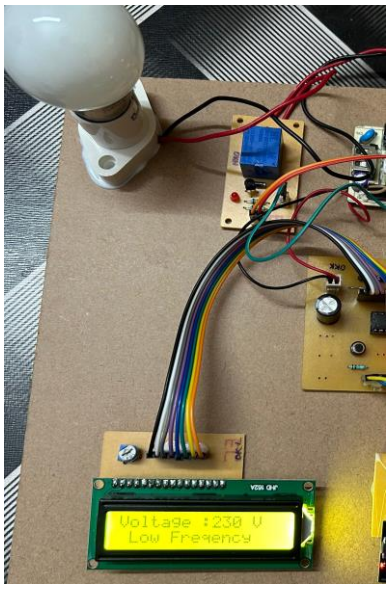


Fig 7.4: Circuit at Normal Voltage and High Frequency

As shown in the Fig.7.4 The level of voltage is 230V and the frequency is 58Hz, those are at Asynchronous condition, so that the bulb is OFF.



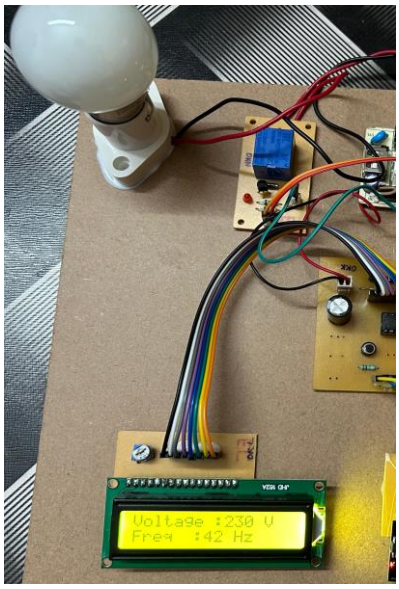


Fig 7.5: Circuit at Normal Voltage and Low Frequency

As shown in the Fig.7.5 The level of voltage is 230V and the frequency is 42Hz, those are at Asynchronous condition, so that the bulb is OFF.

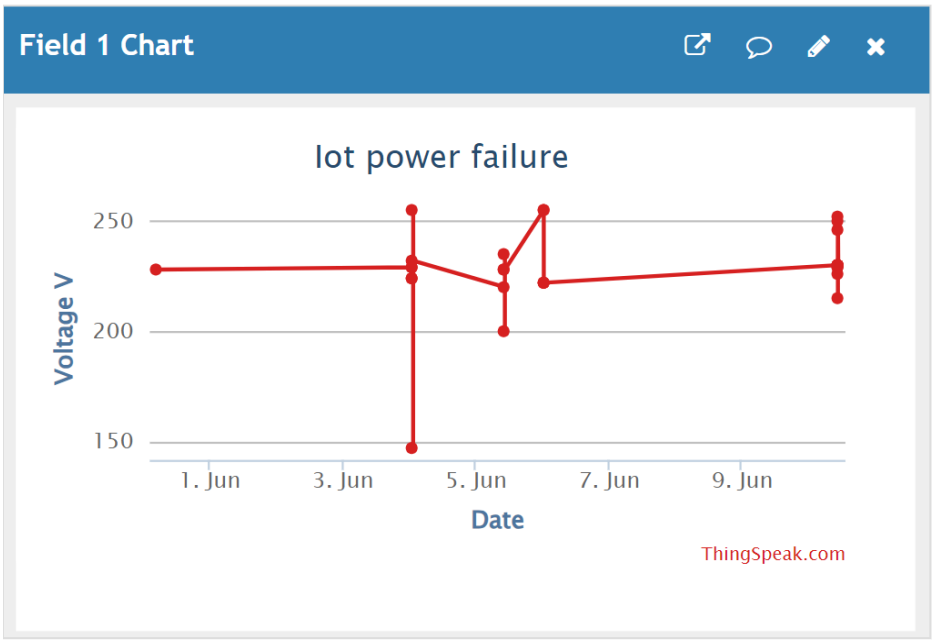


Fig 7.6: The Voltage Profile in ThingSpeak

The Voltage variations are observed in graph using ThingSpeak as shown in the Fig 7.6 which is an IoT platform , such that operator can observe and analyze the values and accordingly adjust to its predetermined values.

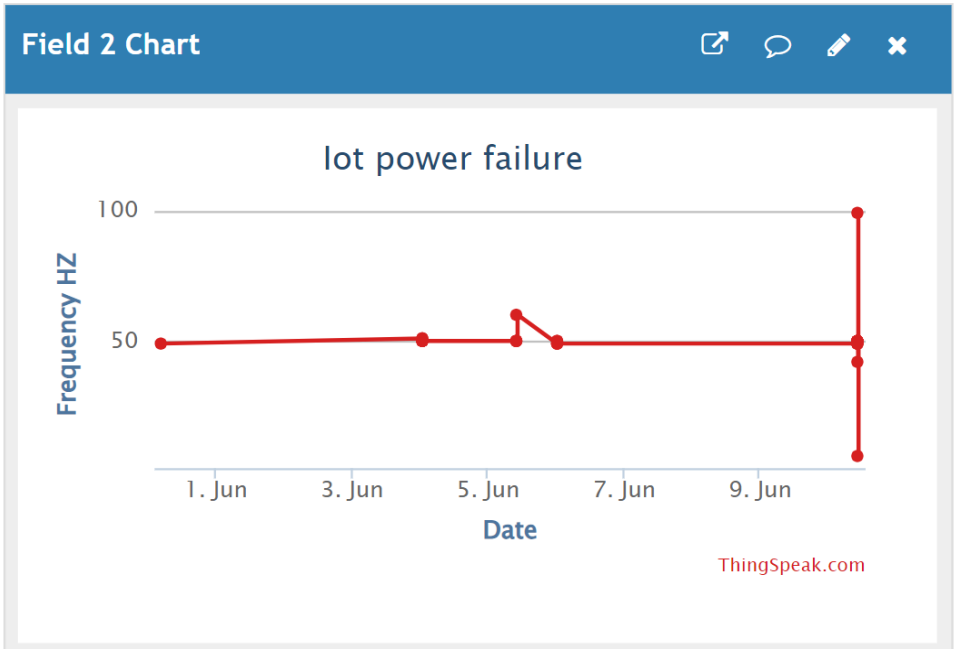


Fig 7.7: The Frequency Profile in ThingSpeak

The Frequency variations are observed in graph using ThingSpeak as shown in the Fig 7.7 which is an IoT platform , such that operator can observe and analyze the values and accordingly adjust to its predetermined values.

7.2 Conclusion:

The goal is to create a system that can sense frequency and voltage deviations to identify when an external power source has lost synchronisation with the power grid. Power comes from a variety of sources , including hydrothermal, solar, and other units connected to the grid. In order to power the load. Grid regulations include a requirement to keep voltage and frequency swings under control. The same feeder must be immediately cut off if the grid's tolerance for variation is exceeded. By detecting fluctuations in voltage and frequency, this prevent widespread power outages. In this project, we'll be using the microcontroller PIC16F73/76 , which, with a bit of tweaking to the code, can do a lot of useful things. In case the Grid completely goes down, it is important to have backup plans ready.

Integrating features of all the hardware components used have been developed in it. Presence of every module has been reasoned out and placed carefully, thus contributing to the best working of the unit. Secondly, using highly advanced IC's with the help of growing technology, the project has been successfully implemented. Thus the project has been successfully designed and tested.

7.3 Future Scope:

In future we can use this project in several applications by adding additional components to this project.

This project can be extended by using GPRS technology, which helps in sending the monitored and controlled data to any place in the world. The temperature controlling systems like coolant can also use in places where temperature level should be maintained.

By connecting wireless camera in industries, factories etc we can see the entire equipments from our personal computer only by using GPRS and GPS technology. The monitoring and controlling of the devices can be done from the personal computer and we can use to handle so many situations.

By connecting temperature sensor, we can get the temperature of dangerous zones in industries and we can use personal computer itself instead of sending human to there and facing problems at the field. The temperature sensor will detect the temperature and it gives information to the micro controller and micro controller gives the information to the mobile phone from that we can get the data at pc side.

Program Code:

The program code which is dumped in the microcontroller of our project is shown below.

```
#include <16F73.h>

#fuses      HS,NOWDT,PROTECT,brownout,put

#use        delay  (clock=20000000)

#include <lcd.c>
#include <ADC.c>

   int16 freq = 0;
   int16 vol = 0;
   int16 count = 0;

void intro()
{
    lcd_putc('\f');
    lcd_gotoxy(1,1);
    printf(lcd_putc,"   WELCOME   ");
    lcd_gotoxy(1,2);
    printf(lcd_putc,"      TO     ");
    delay_ms(2000);
    lcd_putc('\f');
    lcd_gotoxy(1,1);
    printf(lcd_putc,"power grid");
    lcd_gotoxy(1,2);
    printf(lcd_putc,"   failure  monitrng    ");
    delay_ms(2000);
}

void main()
{
```

```
lcd_init();
 delay_ms(500);
 intro();
//Microcontroller Health Check Start

output_high(PIN_C4);
delay_ms(1000);
output_low(PIN_C4);
delay_ms(1000);
output_high(PIN_C4);
delay_ms(1000);
output_low(PIN_C4);
delay_ms(1000);

output_high(PIN_C7);   //Close Relay
output_high(PIN_C4);   // LED Indicator

  set_timer0(0);

 while(1)
 {
   delay_ms(200);

   vol = read_voltage();
   vol = vol();

   freq = 0;
   set_timer0(0);                //Initialise Frequency Counter
   delay_ms(1000);
   freq = get_timer0();          //Get Frequency



   lcd_putc('\f');
   lcd_gotoxy(1,1);
   printf(lcd_putc,"V=%Lu V",vol);
```

```
lcd_gotoxy(1,2);
printf(lcd_putc,"F=%LuHz,T=%Lu C ",freq+1,freq_read);
delay_ms(2000);

if(vol > 240)
{
  high_voltage();
  lcd_putc('\f');
  lcd_gotoxy(1,1);
  printf(lcd_putc," voltage  alert ");
  output_high(pin_C3);
  delay_ms(1000);
}
    if(vol < 170)
{
  output_low(PIN_C7);   //Close Relay
  output_low(PIN_C4);   // LED Indicator

  lcd_putc('\f');
  lcd_gotoxy(1,1);
  printf(lcd_putc,"Volatge alert    ");
  output_high(pin_C3);
  delay_ms(1000);
}

   if(freq_read > 50)
{
  output_low(PIN_C7);   //Close Relay
  output_low(PIN_C4);
          lcd_putc('\f');
  lcd_gotoxy(1,1);
```

```
    printf(lcd_putc,"Frequency alert");
        output_high(pin_C3);
        delay_ms(1000);
     }
          output_low(pin_C3);
   }
}
```

REFERENCES

The sites which were used while doing this project:

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3. www.microchip.com

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3. PCB Design Tutorial –David.L.Jones.

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5. Pyroelectric Sensor Module- Murata.

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